Quantum Computing
Professor Gopal Pandurangan

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Week: 1 (August 24–28, 2026)

Lecture Date: Tuesday, August 25, 2026

Lecture Topic: Efficient Algorithms, Random Sampling, Lower Bound for Deterministic Search, Probability Basics

In this first lecture, we look at two basic problems, primality/factoring and search, introduce random sampling, prove a lower bound of $n-1$ queries for deterministic search, and review the basics of probability.

1 Efficient Algorithms: Primes and Factoring

To decide whether a number n is prime, the input is the binary representation of n, so the *input size* is $N = \log_2 n$ bits. An algorithm is **efficient** if it runs in time polynomial in the input size, here $\mathcal{O}\Big((\log_2 n)^k\Big)$ for a constant k.

Trial division up to $\sqrt{n}$ takes $\Theta(\sqrt{n}) = \Theta\Big(2^{N/2}\Big)$ time, which is exponential. A deterministic polynomial-time algorithm exists ($\tilde{\mathcal{O}}(N^6)$) but is not used in practice. The *randomized* Miller–Rabin test takes $\mathcal{O}(N^3)$ time: if n is composite, at least $3/4$ of the numbers in $\{2, \ldots, n-2\}$ are *witnesses* of this, and a prime has no witness, so after k independent samples a composite n goes undetected with probability at most $(1/4)^k$.

Factoring. Given $n = pq$ with p, q prime, find p and q. Factoring is an *encryption primitive*: if it can be solved quickly, the encryption is broken. No polynomial-time classical algorithm is known, but *Shor's factoring algorithm* is a quantum algorithm with polynomial running time $\mathcal{O}(N^3)$. There is *no* known quantum algorithm that solves NP-complete problems (the hardest problems whose solutions can be verified in polynomial time, such as 3-SAT) in polynomial time.

2 Random Sampling

A randomized algorithm only needs to *sample*. We assume that flipping a (fair) coin is an $\mathcal{O}(1)$ time primitive. Consider an array with n slots (Figure 1), n a power of 2. Every index has $\log n$ bits, so flipping one coin for each bit samples a uniformly random element in $\mathcal{O}(\log n)$ time: $\mathbb{P}(\text{index } i \text{ is chosen}) = (1/2)^{\log n} = 1/n$ for every i.

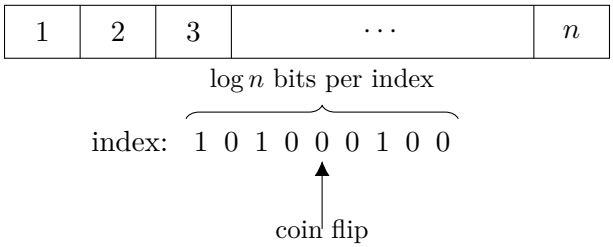


Figure 1: Each of the $\log n$ bits of the index is one coin flip.

Sampling can be done *with* or *without* replacement. We analyze sampling with replacement, since otherwise the samples are not independent. A sample hits a given element with probability $1/n$, so finding it takes $\mathbb{E}[\text{samples}] = \frac{1}{1/n} = n$ samples in expectation.

3 Search

An unsorted array A with n slots can only be accessed through *queries* ("what is $A[i]$?"), and we want the index of a given element x. Linear search makes $\mathcal{O}(n)$ queries. For a lower bound, which has to hold for *all* algorithms, we use an *adversary argument*: the adversary constructs an input on which the given algorithm performs badly.

Theorem 1 ▶ Lower Bound for Deterministic Search

Every deterministic algorithm for search makes at least $n - 1$ queries in the worst case.

Proof. Whenever the algorithm queries a slot i, the adversary answers that $A[i] \neq x$. The algorithm is deterministic, so this defines one run. Suppose that it stops after $k \leqslant n - 2$ queries and outputs an index j. At least 2 slots have not been queried, so the adversary can place x in an unqueried slot different from j. This array is consistent with all the answers, and the algorithm is wrong on it. So a correct algorithm makes at least $n - 1$ queries. □

The argument does not apply to *randomized* algorithms, whose queries are not determined by the previous answers. Randomized search still needs $\Omega(n)$ expected queries, by *Yao's min-max principle*, and Grover's quantum algorithm needs $\mathcal{O}(\sqrt{n})$ queries. To show a *quantum advantage* for a problem, we have to show that it needs $\Omega(f(n))$ queries classically but only $o(f(n))$ queries quantumly.

4 Probability

A **sample space** S is a finite set of *simple events*, $|S| = n$. A **probability function** $\mathbb{P}\colon S \to [0, 1]$ assigns a probability to every simple event. An **event** is a subset $A \subseteq S$, with $\mathbb{P}(A) = \sum_{a \in A} \mathbb{P}(a)$. The *axioms* are $\mathbb{P}(A) \geqslant 0$, $\mathbb{P}(S) = 1$, and $\mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{i=1}^{n} \mathbb{P}(A_i)$ for pairwise disjoint events ($A_i \cap A_j = \emptyset$ for all $i \neq j$). A **random variable** is a function $X\colon S \to V$ to a set of values V, with $\mathbb{P}(X = r) = \sum_{s \in S\,:\,X(s)=r} \mathbb{P}(s)$.

Week: 1 (August 24–28, 2026)

Lecture Date: Thursday, August 27, 2026

Lecture Topic: Search and Satisfiability, Expectation, Bernoulli, Binomial and Geometric Random Variables, Conditional Expectation

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In this lecture, we look at search and satisfiability, define the expectation of a random variable, and compute it for the Bernoulli, binomial and geometric random variables.

1 Search and Satisfiability

A *query* in Grover's algorithm asks for the element at one index of the input.

The *SAT problem* asks for an assignment of the variables of a boolean function $f(x_1, x_2, \ldots, x_n)$ that makes $f = 1$. No polynomial-time algorithm is known. In *3-SAT* the function is given in conjunctive normal form,

$$f = C_1 \wedge C_2 \wedge \cdots \wedge C_m,$$

and a clause such as $C_1 = (x_2 \vee \bar{x}_3)$ contains at most three variables. There is a randomized algorithm for 3-SAT with running time $\mathcal{O}((4/3)^n)$.

SAT is a search over the 2^n assignments, so Grover's algorithm solves it with $\sqrt{2^n} = 2^{n/2} \approx 1.41^n$ queries. In the same way Grover's algorithm applies to every problem that is a search for an input with a given function value.

A quantum advantage has to be shown against randomized algorithms, so we need lower bounds for them and not only for deterministic algorithms.

2 Random Sampling and Expectation

Random sampling is the basis of many randomized algorithms. Unless another distribution is given, the samples are uniform, so one sample from an array of n elements hits a fixed element with probability $1/n$.

Let S be a finite or countable sample space with probability function $\mathbb{P}$, as in the lecture of August 25. For random sampling we take X to be the number of queries until the element is found. Its values are $1, 2, 3, \ldots$ when we sample with replacement and $1, \ldots, n$ when we sample without replacement.

Definition 1 ▶ Expectation

The **expectation** of a random variable X, also called its *first moment*, is

$$\mathbb{E}[X] = \sum_{x \in \text{range}(X)} x \cdot \mathbb{P}(X = x).$$

All random variables in this lecture are nonnegative, so the order of summation in the sums below does not matter.

Theorem 2 ▶ Linearity of Expectation

Let $X = \sum_{i=1}^{n} X_i$, where the X_i are arbitrary random variables. Then

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

Proof. Grouping the simple events s by the value $X(s)$ gives $\mathbb{E}[X] = \sum_{s \in S} X(s)\,\mathbb{P}(s)$. So

$$\mathbb{E}[X] = \sum_{s \in S} \sum_{i=1}^{n} X_i(s)\,\mathbb{P}(s) = \sum_{i=1}^{n} \sum_{s \in S} X_i(s)\,\mathbb{P}(s) = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

□

The X_i do not have to be independent.

Independence. Two events A and B are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$, and **mutually exclusive** if $A \cap B = \emptyset$. The two notions are not the same. Two random variables X and Y are independent if $\mathbb{P}(X = x \text{ and } Y = y) = \mathbb{P}(X = x)\,\mathbb{P}(Y = y)$ for all x and y.

3 Bernoulli, Binomial and Geometric Random Variables

Three random variables appear again and again in the analysis of randomized algorithms.

Definition 2 ▶ Bernoulli Random Variable

A random variable is **Bernoulli** (or 0-1, or an *indicator*) with parameter p if it is 1 with probability p and 0 with probability $1 - p$.

Its expectation is $1 \cdot p + 0 \cdot (1-p) = p$.

Definition 3 ▶ Binomial Random Variable

A **binomial** random variable is a sum of n independent Bernoulli random variables with the same parameter p,

$$B(n,p) = \sum_{i=1}^{n} X_i.$$

It counts the successes among n independent trials with success probability p.

A fixed pattern with k successes and $n-k$ failures has probability $p^k(1-p)^{n-k}$ by independence, and there are $\binom{n}{k}$ such patterns, so $\mathbb{P}(B(n,p)=k) = \binom{n}{k}p^k(1-p)^{n-k}$.

Theorem 3

$\mathbb{E}[B(n,p)] = \sum_{k=0}^{n} k \cdot \mathbb{P}(B(n,p)=k) = np$.

Proof. The first equality is Definition 1. By linearity of expectation (Theorem 2),

$$\mathbb{E}[B(n,p)] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} p = np.$$

□

Definition 4 ▶ Geometric Random Variable

Consider a sequence of independent Bernoulli trials that each succeed with probability $p > 0$. The number of trials up to and including the first success is a **geometric** random variable, $X \sim G(p)$.

The first success is in trial k if the first $k-1$ trials fail and trial k succeeds, so $\mathbb{P}(G(p)=k) = (1-p)^{k-1}p$ and

$$\mathbb{E}[G(p)] = \sum_{k=1}^{\infty} k\,(1-p)^{k-1}p.$$

We compute this sum with conditional expectation.

4 Conditional Probability and Conditional Expectation

Definition 5 ▶ Conditional Probability

For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This is the probability of A once B is known to have happened.

Definition 6 ▶ Conditional Expectation

For random variables X and Y and a value y with $\mathbb{P}(Y = y) > 0$,

$$\mathbb{E}[X \mid Y = y] = \sum_{x \in \text{range}(X)} x \cdot \mathbb{P}(X = x \mid Y = y).$$

The **conditional expectation** $\mathbb{E}[X \mid Y]$ takes the value $\mathbb{E}[X \mid Y = y]$ when $Y = y$.

An expectation is a number, but $\mathbb{E}[X \mid Y]$ is a random variable. It is a function $f(Y)$ of Y.

Theorem 4

$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$.

Proof. Let $f(y) = \mathbb{E}[X \mid Y = y]$. Grouping the simple events by the value of Y, as in the proof of Theorem 2, gives $\mathbb{E}[f(Y)] = \sum_y f(y)\,\mathbb{P}(Y = y)$, where the sums over y are over the values with $\mathbb{P}(Y = y) > 0$. By Definitions 5 and 6,

$$\begin{aligned}\mathbb{E}[\mathbb{E}[X \mid Y]] &= \sum_y \mathbb{P}(Y = y) \sum_x x \cdot \mathbb{P}(X = x \mid Y = y) = \sum_x x \sum_y \mathbb{P}(X = x \text{ and } Y = y) \\ &= \sum_x x \cdot \mathbb{P}(X = x) = \mathbb{E}[X],\end{aligned}$$

because the events $Y = y$ are mutually exclusive and one of them occurs. □

Theorem 5 ▶ Expectation of a Geometric Random Variable

$\mathbb{E}[G(p)] = 1/p$.

Proof. Let $X \sim G(p)$. If $p = 1$, then $X = 1$ always and $\mathbb{E}[X] = 1$. Let $p < 1$, and let $X_1 = 1$ if the first trial succeeds and $X_1 = 0$ otherwise. We condition on X_1. If $X_1 = 1$, then $X = 1$. If $X_1 = 0$, one trial is used and the remaining trials are again a sequence of independent Bernoulli trials, so $\mathbb{E}[X \mid X_1 = 0] = 1 + \mathbb{E}[X]$. By Theorem 4,

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[X \mid X_1 = 1]\,\mathbb{P}(X_1 = 1) + \mathbb{E}[X \mid X_1 = 0]\,\mathbb{P}(X_1 = 0) \\ &= p + (1 - p)(\mathbb{E}[X] + 1) = (1 - p)\,\mathbb{E}[X] + 1.\end{aligned}$$

The sum for $\mathbb{E}[X]$ converges by the ratio test, since consecutive terms have ratio $\frac{(k+1)(1-p)}{k} \to 1 - p < 1$. So $\mathbb{E}[X]$ is finite and we can subtract $(1-p)\,\mathbb{E}[X]$ on both sides. This gives $p\,\mathbb{E}[X] = 1$ and $\mathbb{E}[X] = 1/p$. □

For random sampling with replacement, $X \sim G(1/n)$, so $\mathbb{E}[X] = n$.

Two classes of randomized algorithms. Randomized algorithms are either *Las Vegas* or *Monte Carlo* algorithms (Definition 7). A lower bound for randomized search is shown with Yao's min-max principle (Theorem 8).

Week: 2 (August 31 – September 4, 2026)

Lecture Date: Tuesday, September 1, 2026

Lecture Topic: Query Complexity of Search, Coupon Collector Problem, Lower Bound for Randomized Algorithms, Chernoff Bounds

This lecture covers the query complexity of search for classical randomized algorithms: random sampling, the coupon collector problem, a lower bound by Yao's minmax principle, and Chernoff bounds.

1 Query Complexity for Search

In the search problem of Figure 1, a *query* to a slot x of an array of n slots returns $f(x)$, and the goal is to find the slot x with $f(x) = 1$. The *query complexity* of an algorithm is the number of queries that it makes.

Definition 7 ▶ Monte Carlo and Las Vegas Algorithms

A **Monte Carlo** algorithm is a randomized algorithm whose error probability is > 0.
A **Las Vegas** algorithm is a randomized algorithm whose error probability is $= 0$.

Random sampling *with replacement* queries a slot chosen uniformly at random in every step. Every query succeeds with probability $1/n$, so the number of queries until the first success is a *geometric* random variable with parameter $1/n$, and $\mathbb{E}[\text{queries until find}] = n$ (Theorem 5). Stopped after a fixed number of queries, random sampling is a Monte Carlo algorithm, because it can fail to find the slot. Run until the slot is found, it is a Las Vegas algorithm with n expected queries.

2 Coupon Collector Problem

To distinguish 2 binary arrays with random sampling, all the n slots have to be accessed. The number of queries that this takes is the *coupon collector problem.*

Theorem 6 ▶ Coupon Collector

Let X be the number of queries made by random sampling (with replacement) until all the n slots are accessed. Then

$$\mathbb{E}[X] = nH_n = \Theta(n \log n),$$

where $H_n = \sum_{i=1}^{n} \frac{1}{i}$ is the nth harmonic number.

Proof. Write $X = X_1 + X_2 + \cdots + X_n$, where X_i is the number of queries made to access a new slot after $i-1$ different slots have been accessed. Such a query accesses a new slot with probability $\frac{n-(i-1)}{n}$. So $X_i \sim G\left(\frac{n-(i-1)}{n}\right)$ is geometric with $\mathbb{E}[X_i] = \frac{n}{n-(i-1)}$. By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n \sum_{i=1}^{n} \frac{1}{n-i+1} = nH_n,$$

and $H_n = \Theta(\log n)$. □

2.1 High Probability Bound

Here log is the natural logarithm.

Theorem 7

If random sampling makes $3n \log n$ queries, then

$$\mathbb{P}(\text{fail}) \leqslant \frac{1}{n^2}, \qquad \text{that is,} \qquad \mathbb{P}(\text{Success}) \geqslant 1 - \frac{1}{n^2},$$

where "fail" means that some slot is not accessed.

Proof. Let B_i be the event that slot i is not accessed. A query misses slot i with probability $1 - \frac{1}{n}$, and the queries are independent. Using $1 - x \leqslant e^{-x}$, we get

$$\mathbb{P}(B_i) \leqslant \left(1 - \frac{1}{n}\right)^{3n \log n} \leqslant e^{-3 \log n} = \frac{1}{n^3}, \qquad \mathbb{P}\left(\bigcup_i B_i\right) \leqslant \sum_{i=1}^{n} \mathbb{P}(B_i) \leqslant \frac{n}{n^3} = \frac{1}{n^2}$$

by the union bound. □

3 Lower Bound for Randomized Algorithms

Theorem 8

Any Las Vegas randomized algorithm for search needs $\Omega(n)$ expected queries.

Yao's Minmax Principle

Let A^{rand} be a randomized algorithm, and let $\mathbb{E}\left[A^{\text{rand}}\right]$ be its expected number of queries on its worst input. Let μ be a probability distribution over the inputs, and let D_μ^{best} be the best deterministic algorithm for inputs chosen according to μ. Then

$$\mathbb{E}\left[A^{\text{rand}}\right] \geqslant \max_\mu \mathbb{E}_\mu\left[D_\mu^{\text{best}}\right] \geqslant \Omega(n).$$

The quantity $\mathbb{E}_\mu\left[D_\mu^{\text{best}}\right]$ is called the *distributional complexity.* The first inequality is the principle. The second one holds for search and proves Theorem 8.

Proof of Theorem 8. Let μ choose the slot x with $f(x) = 1$ uniformly at random among the n slots, and let D be a deterministic algorithm that is always correct. As long as all the answers are 0, D queries a fixed sequence of different slots $s_1, s_2, \ldots$, because repeating a query gives no new information. It cannot stop while all the answers are 0 and two slots are not yet queried: either of them could be the solution, and D would be wrong on one of the two inputs. So if the solution is s_j, then D makes at least $\min(j, n-1)$ queries. Every j has probability $1/n$, so

$$\mathbb{E}_\mu[D] \geqslant \frac{1}{n} \sum_{j=1}^{n} \min(j, n-1) \geqslant \frac{1}{n} \sum_{j=1}^{n-1} j = \frac{n-1}{2} = \Omega(n).$$ □

4 Chernoff Bounds

If a coin gives heads with probability $\frac{1}{2} + \varepsilon$ and is tossed n times, then $\mathbb{E}[\text{Heads}] = \frac{n}{2} + n\varepsilon$. Chernoff bounds show that the number of heads is close to its expectation, so $\mathbb{P}(\text{Heads} < \frac{n}{2})$ is very small.

In general, let

$$X = \sum_{i=1}^{n} X_i, \qquad X_i = \begin{cases} 1 & \text{with probability } p_i, \\ 0 & \text{with probability } 1 - p_i, \end{cases}$$

where the X_i are independent Bernoulli random variables. If $p_i = p$ for all i, then $\mathbb{E}[X] = np$ and $\mathbb{V}[X] = np(1-p)$. Figure 2 shows the distribution of X around its mean np.

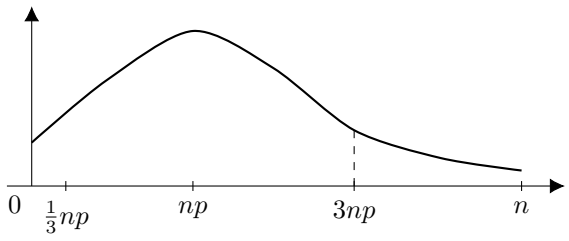


Figure 2: The distribution of X, with mean np.

Theorem 9 ▶ Chernoff Bounds

Let $X = \sum_{i=1}^{n} X_i$ be as above and let $\mu = \mathbb{E}[X]$.

- Upper tail (UI): for $0 < \delta < 1$,

$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{3}}.$$

- For $R \geqslant 6\mu$,

$$\mathbb{P}(X \geqslant R) \leqslant \frac{1}{2^R}.$$

- Lower tail (LI): for $0 < \delta < 1$,

$$\mathbb{P}(X < (1-\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{2}}.$$

4.1 Solution of the Exercise

Exercise 1. Show that if random sampling makes $24n \log n$ queries, then

$$\mathbb{P}(\text{some slot is accessed} < 12 \log n \text{ times}) \leqslant \frac{1}{n^2},$$

so for $n \geqslant 2$ and $c = 6$,

$$\mathbb{P}(\text{some slot is accessed} \leqslant c \log n \text{ times}) \leqslant \frac{1}{n^2}.$$

Proof. Fix a slot i and let $T = 24n \log n$ be the number of queries. Let $Y_i = \sum_{t=1}^{T} Z_t$ be the number of times that slot i is accessed, where $Z_t = 1$ if the tth query accesses slot i and $Z_t = 0$ otherwise. Sampling is with replacement, so the queries are independent, and each accesses slot i with probability $1/n$. So the Z_t are independent Bernoulli random variables with $p_t = 1/n$, as in Theorem 9, and $\mu = \mathbb{E}[Y_i] = T/n = 24 \log n$.
The lower tail bound of Theorem 9 with $\delta = 1/2$, for which $(1-\delta)\mu = 12 \log n$, gives for

every slot i

$$\mathbb{P}(Y_i < 12\log n) \leqslant e^{-\frac{\mu\delta^2}{2}} = e^{-\frac{24\log n}{8}} = \frac{1}{n^3}.$$

By the union bound over the n slots, some slot has $Y_i < 12\log n$ with probability at most $n \cdot 1/n^3 = 1/n^2$. For $n \geqslant 2$ we have $\log n > 0$, so $Y_i \leqslant 6\log n$ implies $Y_i < 12\log n$. So the second event is contained in the first and also has probability at most $1/n^2$. □

The proof uses $24n\log n$ queries and not the $3n\log n$ of Theorem 7. With $\mu = 3\log n$ the lower tail bound is more than $n^{-3/2}$ per slot, which is too weak for the union bound over the n slots. The exercise also holds with "$\geqslant$" in place of "$\leqslant$" and a larger constant ($c = 144$), by the second bound of Theorem 9 with $R = 6\mu$.

Week: 2 (August 31 – September 4, 2026)

Lecture Date: Thursday, September 3, 2026

Lecture Topic: Markov, Chebyshev and Chernoff Inequalities, Search, Birthday Paradox

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In this lecture, we state and prove Markov's inequality and Chebyshev's inequality, and we state the Chernoff bound. We then look at the query complexity of search for classical and quantum algorithms, and we analyze the birthday paradox.

1 Markov's Inequality

Markov's inequality uses only the expectation $\mathbb{E}[X]$ of the random variable, which is also called the *first moment*. Figure 3 shows the bound on the number line.

Theorem 10 ▶ Markov's Inequality

Let $X \geqslant 0$ be a random variable with $\mathbb{E}[X] > 0$. Then, for any $a > 0$,

$$\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}.$$

Equivalently, for any $b > 0$,

$$\mathbb{P}(X \geqslant b) \leqslant \frac{\mathbb{E}[X]}{b}.$$

Proof. Let $I = 1$ if $X \geqslant b$ and $I = 0$ otherwise. Then $I \leqslant \frac{X}{b}$: if $X \geqslant b$ then $I = 1 \leqslant \frac{X}{b}$, and if $X < b$ then $I = 0 \leqslant \frac{X}{b}$ because $X \geqslant 0$. So

$$\mathbb{P}(X \geqslant b) = \mathbb{P}(I = 1) = \mathbb{E}[I] \leqslant \mathbb{E}\left[\frac{X}{b}\right] = \frac{\mathbb{E}[X]}{b}.$$

Taking $b = a\,\mathbb{E}[X] > 0$ gives $\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}$. □

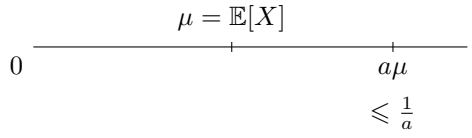


Figure 3: Markov's inequality: X is at least $a\mu$ with probability at most $1/a$.

2 Chebyshev's Inequality and the Chernoff Bound

Chebyshev's inequality uses the variance of the random variable, which is also called the *second moment*.

Definition 8 ▶ Variance and Standard Deviation

The **variance** of a random variable X is $\mathbb{V}[X] = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]$. The **standard deviation** of X is $\sigma = \sqrt{\mathbb{V}[X]}$.

Theorem 11 ▶ Chebyshev's Inequality

For any random variable X and any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) \leqslant \frac{\mathbb{V}[X]}{a^2} = \frac{\sigma^2}{a^2}.$$

Proof.

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) = \mathbb{P}\Big((X - \mathbb{E}[X])^2 \geqslant a^2\Big) \leqslant \frac{\mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]}{a^2} = \frac{\mathbb{V}[X]}{a^2},$$

by Markov's inequality (Theorem 10) for $(X - \mathbb{E}[X])^2 \geqslant 0$. □

Chernoff bound. The Chernoff bounds (Theorem 9) need $X = \sum_{i=1}^{n} X_i$ to be a sum of *mutually independent* indicator random variables. Then $\mu = \mathbb{E}[X] = \sum_{i=1}^{n} p_i$ with $p_i = \mathbb{P}(X_i = 1)$ by linearity of expectation.

3 Search

Let $f\colon D \to R$ be a function from a domain D to a range R. Given y, we ask: is there an x such that $f(x) = y$ (Figure 4)? We consider $f\colon [1, \ldots, N] \to [1, \ldots, N]$ and want to find an x such that $f(x) = k$. The function is an array with n slots (Figure 1), where n is the size of the domain, and the cost of an algorithm is the number of queries to f. A randomized classical algorithm needs $\Theta(n)$ queries, and Grover's algorithm needs $\mathcal{O}(\sqrt{n})$ queries. For the random sampling algorithm, $\mathbb{E}[\text{queries}] = \mathcal{O}(n)$ and one sample has $\mathbb{P}(\text{Success}) = 1/n$.

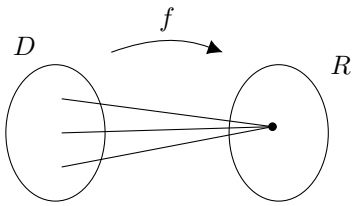


Figure 4: A function $f\colon D \to R$; several x can have the same value $y = f(x)$.

Probability amplitude amplification. If the classical success probability is p, then in the quantum case it becomes $\sqrt{p}$.

4 Birthday Paradox

There are m people and n birthdays ($n = 365$), and $f(x)$ is the birthday of person x. A *collision* is a pair $x \neq y$ with $f(x) = f(y)$.

Let E_i be the event that the ith birthday is different from the first $i-1$ birthdays. By the chain rule and $1 - x \leqslant e^{-x}$,

$$\mathbb{P}(\text{all birthdays different}) = \mathbb{P}(E_1) \cdot \mathbb{P}(E_2 \mid E_1) \cdot \mathbb{P}(E_3 \mid E_2 \cap E_1) \cdots = \prod_{i=0}^{m-1} \left(1 - \frac{i}{n}\right)$$
$$\leqslant \prod_{i=0}^{m-1} e^{-i/n} = e^{-\frac{(m-1)m}{2n}}.$$

For $m = \sqrt{2n} + 1$ we have $(m-1)m \geqslant 2n$, so this is at most $1/e \leqslant 1/2$. So $\mathcal{O}(\sqrt{n})$ people are enough to have a constant probability of collision.

4.1 Expected Number of Collisions

The birthday paradox can be seen as throwing m balls into n bins (Figure 5). For $m \geqslant \sqrt{2n}+1$, $\mathbb{E}[\#\text{ collisions}] \geqslant 1$.

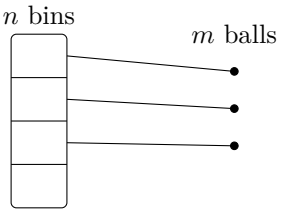


Figure 5: Throwing m balls into n bins.

Let X be the number of collisions. X takes values from 0 to $\binom{m}{2}$. Write

$$X = \sum_{i \neq j} X_{ij}, \qquad X_{ij} = \begin{cases} 1 & \text{if } f(i) = f(j), \\ 0 & \text{else,} \end{cases}$$

where the sum is over the $\binom{m}{2}$ pairs $i \neq j$. By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i \neq j} \mathbb{P}(X_{ij} = 1) = \binom{m}{2} \frac{1}{n} = \frac{m(m-1)}{2n} \sim \frac{m^2}{2n} = \Theta\left(\frac{m^2}{n}\right).$$

For $m \geqslant \sqrt{2n} + 1$ we have $m(m-1) \geqslant 2n$, so $\mathbb{E}[X] \geqslant 1$.

Lemma 12 ▶ Variance of the Number of Collisions

The X_{ij} are *pairwise independent*: for any two different pairs $\{i, j\} \neq \{k, l\}$,

$$\mathbb{E}[X_{ij} X_{kl}] = \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}] = \frac{1}{n^2}.$$

So

$$\mathbb{V}[X] = \sum_{i \neq j} \mathbb{V}[X_{ij}] = \binom{m}{2} \frac{1}{n}\left(1 - \frac{1}{n}\right) \leqslant \mathbb{E}[X].$$

Proof. We have $\mathbb{E}[X_{ij}X_{kl}] = \mathbb{P}(f(i) = f(j) \text{ and } f(k) = f(l))$. If the pairs are disjoint, the two events depend on the independent birthdays of four different people, so the probability is $\mathbb{P}(f(i) = f(j)) \cdot \mathbb{P}(f(k) = f(l)) = 1/n^2$. Otherwise the pairs share one person, say $\{i, j\}$ and $\{i, k\}$ with $j \neq k$. Then $f(j)$ and $f(k)$ are independent of $f(i)$ and of each other, and each equals a given day b with probability $1/n$, so

$$\mathbb{P}(f(j) = f(i) \text{ and } f(k) = f(i)) = \sum_{b=1}^{n} \mathbb{P}(f(i) = b) \cdot \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n^2}.$$

In both cases $\mathbb{E}[X_{ij}X_{kl}] = 1/n^2 = \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]$. For two indicators this is the same as independence, because their joint distribution is fixed by $\mathbb{P}(X_{ij} = 1)$, $\mathbb{P}(X_{kl} = 1)$ and $\mathbb{P}(X_{ij} = X_{kl} = 1)$.

Expand the square of $X - \mathbb{E}[X] = \sum_{i \neq j}(X_{ij} - \mathbb{E}[X_{ij}])$:

$$\begin{aligned}\mathbb{V}[X] &= \sum_{i \neq j} \mathbb{E}\Big[(X_{ij} - \mathbb{E}[X_{ij}])^2\Big] + \sum_{\{i,j\} \neq \{k,l\}} \mathbb{E}[(X_{ij} - \mathbb{E}[X_{ij}])(X_{kl} - \mathbb{E}[X_{kl}])] \\ &= \sum_{i \neq j} \mathbb{V}[X_{ij}] + \sum_{\{i,j\} \neq \{k,l\}} (\mathbb{E}[X_{ij}X_{kl}] - \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]) = \sum_{i \neq j} \mathbb{V}[X_{ij}],\end{aligned}$$

because the second sum is zero by the first part. Since $X_{ij}^2 = X_{ij}$, $\mathbb{V}[X_{ij}] = \mathbb{E}\Big[X_{ij}^2\Big] - \mathbb{E}[X_{ij}]^2 = \frac{1}{n}\Big(1 - \frac{1}{n}\Big) \leqslant \frac{1}{n}$. Summing over the $\binom{m}{2}$ pairs gives $\mathbb{V}[X] \leqslant \binom{m}{2}\frac{1}{n} = \mathbb{E}[X]$. □

Exercise 2. Show that the birthday paradox occurs: with $\mathcal{O}(\sqrt{n})$ people, $\mathbb{P}(X \geqslant 1) \geqslant \Theta(1)$. Precisely, if $m \geqslant 2\sqrt{n} + 1$, then

$$\mathbb{P}(X \geqslant 1) \geqslant \frac{1}{2}.$$

Proof. If $X = 0$ then $|X - \mathbb{E}[X]| = \mathbb{E}[X]$. So by Chebyshev's inequality (Theorem 11) with $a = \mathbb{E}[X]$, and then Lemma 12,

$$\mathbb{P}(X = 0) \leqslant \mathbb{P}(|X - \mathbb{E}[X]| \geqslant \mathbb{E}[X]) \leqslant \frac{\mathbb{V}[X]}{\mathbb{E}[X]^2} \leqslant \frac{1}{\mathbb{E}[X]}.$$

For $m \geqslant 2\sqrt{n} + 1$, since $m(m-1)$ grows with m, $\mathbb{E}[X] = \frac{m(m-1)}{2n} \geqslant \frac{(2\sqrt{n}+1)\cdot 2\sqrt{n}}{2n} \geqslant 2$, so $\mathbb{P}(X = 0) \leqslant 1/2$ and $\mathbb{P}(X \geqslant 1) = 1 - \mathbb{P}(X = 0) \geqslant 1/2$. □

5 Quantum Birthday Paradox

The *collision problem* is: given query access to a function f, find $x \neq y$ with $f(x) = f(y)$. By Exercise 2, $\mathcal{O}(\sqrt{n})$ classical queries to random points are enough. An algorithm makes k queries $f(x_1), \ldots, f(x_k)$, where the point x_s may depend on the earlier answers $f(x_1), \ldots, f(x_{s-1})$ and on random coins.

As in the birthday paradox, $f\colon D \to [n]$, where $[n] = \{1, \ldots, n\}$, is a *random function*: every value $f(x)$ is uniform in $[n]$ and independent of all the other values. The algorithm *finds a collision* if it outputs $x \neq y$ with $f(x) = f(y)$. We may assume that it queries the two points that it outputs. This costs at most 2 extra queries, which does not affect the bound $\Omega(\sqrt{n})$. We may also assume that the queried points are different, because repeating a query gives no new information.

Exercise 3. Show a $\sqrt{n}$ lower bound in the classical case: for a random function $f\colon D \to [n]$, every classical algorithm (deterministic or randomized, adaptive) that makes k queries to different points, including the two points that it outputs, finds a collision with probability

at most

$$\frac{\binom{k}{2}}{n} = \frac{k(k-1)}{2n}.$$

So to find a collision with probability at least $1/2$ it needs $k > \sqrt{n}$ queries, and the classical query complexity of the collision problem is $\Omega(\sqrt{n})$.

Proof. The algorithm queries the points that it outputs, so it can only find a collision if two of its queried values agree. Let E_s be the event that $f(x_s) \in \{f(x_1), \ldots, f(x_{s-1})\}$. A collision needs E_s to occur for some s. Condition on the first $s-1$ answers and on the random coins of the algorithm. Then x_s is a fixed point that has not been queried before. So $f(x_s)$ is still uniform in $[n]$ and independent of the $s-1$ values $f(x_1), \ldots, f(x_{s-1})$, which are now fixed. It equals one of them with probability at most $\frac{s-1}{n}$, no matter how the algorithm chose x_s. This holds for every outcome of the first $s-1$ steps and of the coins, so $\mathbb{P}(E_s) \leqslant \frac{s-1}{n}$ without conditioning. By the union bound,

$$\mathbb{P}(\text{collision}) \leqslant \sum_{s=1}^{k} \mathbb{P}(E_s) \leqslant \sum_{s=1}^{k} \frac{s-1}{n} = \frac{k(k-1)}{2n}.$$

If the algorithm succeeds with probability at least $1/2$, then $\frac{k(k-1)}{2n} \geqslant \frac{1}{2}$, so $k^2 > k(k-1) \geqslant n$ and $k > \sqrt{n}$. □

Week: 3 (September 7–11, 2026)

Lecture Date: Tuesday, September 8, 2026

Lecture Topic: Distributed Computing, Randomized Leader Election in a Complete Network, Random Walks, Law of Large Numbers

We define the synchronous complete network model and analyze a randomized algorithm that elects a leader in $\mathcal{O}(1)$ rounds with $\mathcal{O}\left(\sqrt{n}\log^{3/2} n\right)$ messages. We then define random walks on undirected graphs and prove the weak law of large numbers with Chebyshev's inequality.

1 Distributed Computing in a Complete Network

The network is a *complete graph* G on n nodes (Figure 6). Every pair of nodes is joined by a bidirectional communication link, so there are $\binom{n}{2} = \frac{n(n-1)}{2}$ edges and every node has $n-1$ incident edges. The network is *synchronous*: all nodes share a clock, and the computation runs in *rounds*. In one round, every node receives the messages sent to it in the previous round, does a local computation, and sends messages on some of its edges (receives $\to$ computes $\to$ sends).

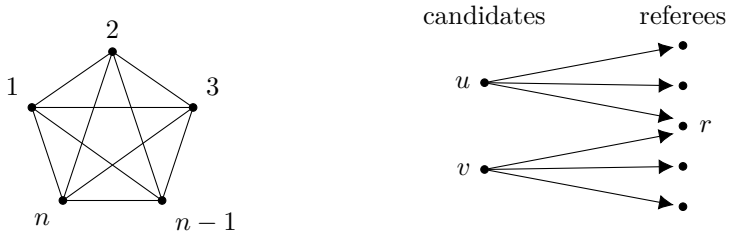


Figure 6: Left: the complete network on n nodes. Right: candidates u and v with a common referee r.

1.1 Leader Election

Problem 1 ▶ Leader Election

All nodes run the same algorithm. At the end, exactly one node must be in the state *leader*, and all other nodes must be in the state *non-leader*.

The goals are to minimize the number of rounds and the total number of messages sent. The nodes are either *labelled* (every node has a unique id) or *anonymous* (no ids).

The trivial algorithm. With unique ids, every node sends its id to all other nodes in the first round, and the node with the minimum id becomes the leader. This takes $\mathcal{O}(1)$ rounds and $n(n-1) = \Theta(n^2)$ messages.

Anonymous nodes. A node addresses its edges by $n-1$ *ports*, numbered $1, \ldots, n-1$, one per incident edge. It can send a message on a given port and sees on which port a message arrives. It does not know which node is at the other end of a port, and the port numbering is chosen by an adversary.

Exercise 4. Let $n \geqslant 2$. Prove that in an anonymous complete network in which all nodes start in the same state, there is no deterministic algorithm that elects a leader for every port numbering: for every deterministic algorithm there is a port numbering for which, after every number of rounds, the number of leaders is either 0 or n.

Proof. Fix a deterministic algorithm. Number the nodes by $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ (addition modulo n), and let port j of node i lead to node $i + j$, for $j = 1, \ldots, n-1$. This is a valid port numbering, since $i+1, \ldots, i+(n-1)$ are exactly the nodes other than i. The edge between i and $i + j$ is port j at i and port $n - j$ at $i + j$, since $(i + j) + (n - j) = i$ in $\mathbb{Z}_n$. So a message sent on port j arrives on port $n - j$.

We show by induction on t that at the end of round t all nodes are in the same state, and that in round t all nodes send the same message m_j on port j, for every j ("no message" counts as a message). For $t = 0$ all nodes are in the same initial state, and nothing has been sent. If the claim holds for round $t - 1$, then in round t node i receives on port q the message that node $i + q$ sent on its port $n - q$, which is the common message m_{n-q} of round $t - 1$ and does not depend on i. So all nodes are in the same state and receive the same messages on the same ports. The algorithm is deterministic, so they compute the same new state and send the same messages on the same ports.

Whether a node declares itself the leader depends only on its state. At the end of every round, either all $n \geqslant 2$ nodes are leaders or no node is, so there is never exactly one leader. □

From now on we assume that the nodes have unique ids.

Complexity classes. Deterministic, bounded-error probabilistic and bounded-error quantum polynomial time satisfy $\mathrm{P} \subseteq \mathrm{BPP} \subseteq \mathrm{BQP}$.

1.2 Known Bounds

For leader election in a complete network of n nodes with unique ids, the following bounds are known ($\widetilde{O}$ and $\widetilde{\Omega}$ hide factors polynomial in $\log n$). We prove only the first and the third row.

Messages	Rounds	Algorithm
$\Theta(n^2)$	$\mathcal{O}(1)$	trivial, deterministic
$\mathcal{O}(n \log n)$	$\mathcal{O}(\log n)$	deterministic
$\mathcal{O}\left(\sqrt{n} \log^{3/2} n\right) = \widetilde{O}(\sqrt{n})$	$\mathcal{O}(1)$	randomized (this lecture)
$\widetilde{\Omega}(\sqrt{n})$		classical lower bound
$\widetilde{O}(n^{1/3})$		quantum

2 A Randomized Algorithm for Leader Election

In this section, log is the natural logarithm, $p = \frac{2 \log n}{n}$ and $k = \lceil \sqrt{n \log n} \rceil$. We have $p \leqslant 1$ and $k \leqslant n$ for every $n \geqslant 1$, since $2 \log n \leqslant n$.

1. **Subsample candidates.** Every node becomes a *candidate* with probability p, independently of all other nodes.
2. **Candidates contact referees.** Every candidate sends its id to k random nodes, its *referees*.
3. **Referee round.** Every node that received at least one id declares the minimum id among the received messages the *winner*, and replies with the winner's id to all candidates that contacted it.

4. **Decision.** A candidate becomes the leader if it is the winner at all of its referees. All other nodes are non-leaders.

The model for the random choices. Every node u draws a coin $C_u \in \{0,1\}$ with $\mathbb{P}(C_u = 1) = p$ (u is a candidate if $C_u = 1$) and a *referee set* R_u, which is a uniformly random set of k distinct nodes among all n nodes. All the C_u and all the R_u are mutually independent. Only candidates use their referee sets. A candidate may pick itself as a referee. It then plays the role of this referee itself, and no message is needed.

2.1 The Number of Candidates

Let $X = \sum_u C_u$ be the number of candidates. By linearity of expectation, $\mathbb{E}[X] = np = 2\log n$.

Exercise 5. Show that there is at least one leader if and only if there is at least one candidate, and that

$$\mathbb{P}(\text{no leader}) = \mathbb{P}(X = 0) \leqslant \frac{1}{n^2}, \qquad \text{so} \qquad \mathbb{P}(\text{at least 1 leader}) \geqslant 1 - \frac{1}{n^2}.$$

Proof. Only candidates become leaders, so if $X = 0$ there is no leader. If $X \geqslant 1$, let w be the candidate with the minimum id (unique, since the ids are unique). Every referee r of w receives only ids of candidates. All of them are at least the id of w, and the id of w is one of them. So r declares w the winner, and w is a leader. The coins are independent, so, using $0 \leqslant 1 - p$ and $1 - x \leqslant e^{-x}$, $\mathbb{P}(X = 0) = (1-p)^n \leqslant e^{-pn} = e^{-2\log n} = 1/n^2$. □

2.2 At Most One Leader

A pair of different nodes $\{u, v\}$ is a *bad pair* if both are candidates and they have no common referee: $C_u = C_v = 1$ and $R_u \cap R_v = \emptyset$.

Lemma 13 ▶ Two Leaders Form a Bad Pair

If two candidates have a common referee, then at most one of them is a leader. So if there is more than one leader, then there is a bad pair.

Proof. Let $u \neq v$ be candidates with a common referee r, where the id of u is smaller than the id of v (Figure 6). The referee r receives both ids (if r is u or v itself, it knows its own id), so the minimum id that it receives is at most the id of u. Then r does not declare v the winner, and v is not a leader. So two different leaders are two candidates without a common referee, which is a bad pair. □

Lemma 14 ▶ Two Referee Sets Rarely Miss Each Other

For any two different nodes u and v,

$$\mathbb{P}(R_u \cap R_v = \emptyset) \leqslant \left(1 - \frac{k}{n}\right)^k \leqslant e^{-k^2/n} \leqslant \frac{1}{n}.$$

Proof. Condition on $R_u = S$, where S is any set of k nodes. By independence, R_v is still a uniformly random set of k nodes. If $2k > n$, two sets of k nodes cannot be disjoint and the probability is 0. So let $2k \leqslant n$. Draw the k nodes of R_v one after the other, each uniformly

among the nodes not drawn yet. If the first i nodes drawn are outside S, then $n-i$ nodes are left and k of them are in S, so the next node is outside S with probability $\frac{n-k-i}{n-i}$. So

$$\mathbb{P}(R_v \cap S = \emptyset) = \prod_{i=0}^{k-1}\left(1 - \frac{k}{n-i}\right) \leqslant \left(1 - \frac{k}{n}\right)^k \leqslant e^{-k^2/n} \leqslant e^{-\log n} = \frac{1}{n},$$

because every factor satisfies $0 \leqslant 1 - \frac{k}{n-i} \leqslant 1 - \frac{k}{n}$ (since $i \leqslant k-1$ and $2k \leqslant n$ give $n - i \geqslant k$), and by $1 - x \leqslant e^{-x}$ and $k^2 \geqslant n \log n$. The bound does not depend on S, so it also holds without conditioning. □

Exercise 6. Show that

$$\mathbb{P}(\text{more than 1 leader}) \leqslant \frac{2\log^2 n}{n},$$

so there is at most 1 leader with probability tending to 1.

Proof. Let B be the number of bad pairs. For a fixed pair $\{u, v\}$, the coins C_u, C_v and the pair of sets (R_u, R_v) are independent, so by Lemma 14 the pair is bad with probability $p^2 \cdot \mathbb{P}(R_u \cap R_v = \emptyset) \leqslant p^2/n$. By linearity of expectation over the $\binom{n}{2}$ pairs, $\mathbb{E}[B] \leqslant \binom{n}{2}\frac{p^2}{n} = \frac{n(n-1)}{2} \cdot \frac{4\log^2 n}{n^3} \leqslant \frac{2\log^2 n}{n}$. If there is more than one leader, then $B \geqslant 1$ by Lemma 13. Since $B \geqslant 0$, Markov's inequality (Theorem 10) with $b = 1$ gives $\mathbb{P}(\text{more than 1 leader}) \leqslant \mathbb{P}(B \geqslant 1) \leqslant \mathbb{E}[B]$, and $2\log^2 n/n \to 0$ as $n \to \infty$. □

2.3 Messages and Rounds

Theorem 15 ▶ Complexity of the Algorithm

The algorithm takes 3 rounds, and messages are sent only in the first two. The expected number of messages is at most $4k \log n = \mathcal{O}\left(\sqrt{n}\log^{3/2} n\right)$.

Proof. Rounds. In round 1 every node draws its coin and its referee set, and the candidates send their ids. In round 2 the referees receive the ids and reply with the minimum. In round 3 the candidates receive the replies and decide, and no message is sent.

Messages. Every candidate sends at most k messages and gets at most k replies, so the number of messages is at most $2kX$. Its expectation is at most $2k\,\mathbb{E}[X] = 4k \log n$, which is $\mathcal{O}\left(\sqrt{n}\log^{3/2} n\right)$ since $k \leqslant \sqrt{n \log n} + 1$. □

3 Markov Chains: Random Walks in Undirected Graphs

Definition 9 ▶ Random Walk

Let G be an undirected graph in which every node has at least one neighbour. In a **random walk** on G, a walker starts at some node. In every step it moves from its current node u to each neighbour of u with probability $\frac{1}{\deg(u)}$, independently of the earlier steps.

Let $p_t = (p_t^1, \ldots, p_t^n)$ be the row vector where p_t^i is the probability that the walker is at node i after t steps. The walker is at one node, so the entries of p_t sum to 1. If the walker is at node j after t steps, it moves to a given neighbour i of j with probability $\frac{1}{\deg(j)}$. So by the law of

total probability,

$$p_{t+1}^i = \sum_{j\,:\,j \text{ is a neighbour of } i} \frac{p_t^j}{\deg(j)}. \tag{1}$$

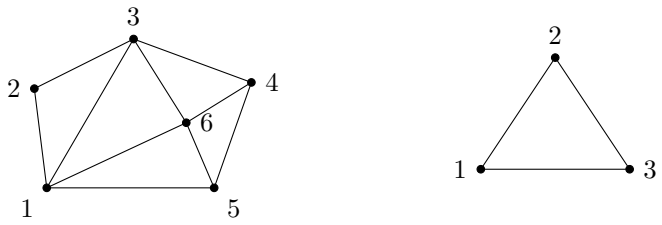


Figure 7: Left: a graph with 6 nodes and 10 edges. Right: the triangle.

Example 1. In the graph on the left of Figure 7, a walker that starts at node 1 has $p_0 = (1,0,0,0,0,0)$ and, since node 1 has the 4 neighbours 2, 3, 5, 6, $p_1 = \left(0, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}\right)$. In the triangle on the right every node has degree 2, and starting at node 1, by Equation (1), $p_0 = (1,0,0)$, $p_1 = \left(0, \frac{1}{2}, \frac{1}{2}\right)$ and $p_2 = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$.

4 The Law of Large Numbers

Exercise 7. Let $X_1, X_2, \ldots$ be pairwise independent random variables with the same expectation $\mathbb{E}[X_i] = \mu$ and the same finite variance $\mathbb{V}[X_i] = \sigma^2$, and let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. Show the weak law of large numbers with Chebyshev's inequality: for every $\varepsilon > 0$,

$$\mathbb{P}\Big(\Big|\overline{X}_n - \mu\Big| \geqslant \varepsilon\Big) \leqslant \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Proof. By linearity of expectation, $\mathbb{E}\Big[\overline{X}_n\Big] = \frac{1}{n}\sum_i \mathbb{E}[X_i] = \mu$. Let $Y_i = X_i - \mu$. Then $\mathbb{E}[Y_i] = 0$, $\mathbb{E}\big[Y_i^2\big] = \sigma^2$, and $\overline{X}_n - \mu = \frac{1}{n}\sum_i Y_i$. For $i \neq j$, Y_i and Y_j are independent, so $\mathbb{E}[Y_iY_j] = \mathbb{E}[Y_i]\,\mathbb{E}[Y_j] = 0$. Expanding the square as in the proof of Lemma 12 gives

$$\mathbb{V}\Big[\overline{X}_n\Big] = \mathbb{E}\left[\Big(\frac{1}{n}\sum_{i=1}^n Y_i\Big)^2\right] = \frac{1}{n^2}\sum_{i=1}^n \mathbb{E}\Big[Y_i^2\Big] + \frac{1}{n^2}\sum_{i\neq j}\mathbb{E}[Y_iY_j] = \frac{\sigma^2}{n}.$$

Chebyshev's inequality (Theorem 11) for $\overline{X}_n$ with $a = \varepsilon$ gives $\mathbb{P}\Big(\Big|\overline{X}_n - \mu\Big| \geqslant \varepsilon\Big) \leqslant \mathbb{V}\Big[\overline{X}_n\Big]/\varepsilon^2 = \sigma^2/(n\varepsilon^2)$, which tends to 0 as $n \to \infty$. □

Week: 3 (September 7–11, 2026)

Lecture Date: Thursday, September 10, 2026

Lecture Topic: Law of Large Numbers, Random Walks on Graphs, Transition Matrix, Stationary Distribution and Convergence

We state the law of large numbers. For a random walk on a graph we define the transition matrix P, show that $p_{t+1} = p_t P$, find the stationary distribution, and state the convergence theorem.

Law of large numbers. We proved the weak form in Exercise 7.

Theorem 16 ▶ Law of Large Numbers

Let $X_1, X_2, \dots$ be independent, identically distributed (i.i.d.) random variables distributed as X, where $\mathbb{E}[X]$ is finite. Then, with probability 1,

$$\lim_{n\to\infty} \frac{\sum_{i=1}^{n} X_i}{n} = \mathbb{E}[X].$$

1 Random Walks on Graphs

Let $G = (V, E)$ be a finite undirected graph without isolated vertices, with $V = \{1, \dots, n\}$ and $m = |E|$ edges. We write $u \sim v$ if $\{u, v\} \in E$, and $\deg(u)$ (or d_u) for the number of neighbours of u. The walker is now called the token, and X_t is its vertex at time $t = 0, 1, 2, \dots$.

Definition 10 ▶ Distribution at Time t, Transition Matrix

The **distribution at time** t is the row vector

$$p_t = \left(p_t^1, p_t^2, \dots, p_t^n\right), \qquad p_t^i = \mathbb{P}(X_t = i).$$

The **transition matrix** is the $n \times n$ matrix P with entries

$$P_{uv} = \mathbb{P}(u \to v) = \mathbb{P}(X_{t+1} = v \mid X_t = u) = \begin{cases} \frac{1}{\deg(u)} & \text{if } u \sim v, \\ 0 & \text{otherwise.} \end{cases}$$

1.1 The Triangle

In the triangle on the vertices $1, 2, 3$ (the right half of Figure 7) every vertex has degree 2, so P is the 3×3 matrix with 0 on the diagonal and $1/2$ in every other entry.

1.2 The Evolution $p_{t+1} = p_t P$

Lemma 17 ▶ Evolution of the Distribution

For every $t \geqslant 0$,

$$p_{t+1} = p_t P, \qquad \text{and so} \qquad p_t = p_0 P^t.$$

Proof. Fix a vertex v. The events $\{X_t = u\}$, $u \in V$, are disjoint and one of them occurs, so

$$p_{t+1}^v = \mathbb{P}(X_{t+1} = v) = \sum_{u \in V} \mathbb{P}(X_t = u \text{ and } X_{t+1} = v) = \sum_{u \in V} p_t^u P_{uv} = (p_t P)_v.$$

Here $\mathbb{P}(X_t = u \text{ and } X_{t+1} = v) = p_t^u P_{uv}$ by the definition of P_{uv} as a conditional probability when $p_t^u > 0$, and both sides are 0 when $p_t^u = 0$. So $p_{t+1} = p_t P$. Induction on t (with $P^0 = I$) gives $p_t = p_0 P^t$. □

Entry by entry, $p_{t+1} = p_t P$ is Equation (1). For the triangle with $p_0 = (1, 0, 0)$, p_1 and p_2 are in Example 1.

1.3 Row Stochastic Matrices

Definition 11 ▶ Probability Vector, Row Stochastic Matrix

A vector $p = (p^1, \ldots, p^n)$ is a **probability vector** if $p^u \geqslant 0$ for all u and $\sum_{u=1}^n p^u = 1$. An $n \times n$ matrix P is **row stochastic** if every row of P is a probability vector:

$$P_{uv} \geqslant 0 \text{ for all } u, v, \qquad \text{and} \qquad \sum_{v=1}^{n} P_{uv} = 1 \text{ for all } u.$$

The transition matrix of a random walk on any graph is row stochastic: row u has $\deg(u)$ entries equal to $1/\deg(u)$, and its other entries are 0.

Exercise 8. Let P be a row stochastic $n \times n$ matrix and let p be a probability vector. Show that pP is a probability vector. So if p_0 is a probability vector and $p_{t+1} = p_t P$ for all $t \geqslant 0$, then $p_t = p_0 P^t$ is a probability vector for every $t \geqslant 0$.

Proof. Let $q = pP$, so $q^v = \sum_{u=1}^n p^u P_{uv}$ for every v. Every term is a product of two nonnegative numbers, so $q^v \geqslant 0$. Every row of P and the entries of p sum to 1, so $\sum_v q^v = \sum_v \sum_u p^u P_{uv} = \sum_u p^u (\sum_v P_{uv}) = \sum_u p^u = 1$. So pP is a probability vector. The vector p_0 is one by assumption, and if p_t is one, then so is $p_t P = p_{t+1}$. By induction on t, every p_t is a probability vector. □

1.4 The Transition Matrix of a General Graph

Let A be the *adjacency matrix* of G, with $A_{uv} = 1$ if $u \sim v$ and $A_{uv} = 0$ otherwise. Let D be the diagonal matrix with diagonal entries $d_1, \ldots, d_n$. Its inverse D^{-1} has diagonal entries $1/d_1, \ldots, 1/d_n$ and exists because there are no isolated vertices. Then $(D^{-1}A)_{uv} = A_{uv}/\deg(u)$, which is P_{uv} of Definition 10, so $P = D^{-1}A$. If G is *d-regular*, meaning every vertex has the same degree d, then $D = dI$ and $P = A/d$. The triangle is 2-regular, and its P is $A/2$.

2 Stationary Distribution

Definition 12 ▶ Stationary Distribution

A probability vector π is a **stationary distribution** of the walk with transition matrix P if

$$\pi P = \pi \cdot 1 = \pi.$$

Then π is a left eigenvector of P with eigenvalue 1.

By Lemma 17, if $p_t = \pi$ then $p_{t+1} = \pi P = \pi$. Only the neighbours v of u have $P_{vu} \neq 0$, and for them $P_{vu} = 1/\deg(v)$, so (as in Equation (1))

$$(\pi P)_u = \sum_{v \in V} \pi_v P_{vu} = \sum_{v \sim u} \frac{\pi_v}{\deg(v)}. \tag{2}$$

The sum runs over the neighbours of u, as in Figure 8.

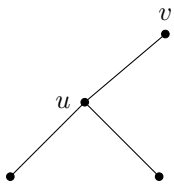


Figure 8: A vertex u and its neighbours; v is one of them.

Theorem 18 ▶ Stationary Distribution of a Random Walk on a Graph

Let G be a graph with m edges and no isolated vertices. Then

$$\pi_v = \frac{\deg(v)}{2m} \qquad (v \in V)$$

is a stationary distribution of the random walk on G.

Proof. π is a probability vector. There are no isolated vertices and V is not empty, so $m \geqslant 1$. So π is well defined, and $\pi_v \geqslant 0$. Every edge $\{u, v\}$ is counted once in $\deg(u)$ and once in $\deg(v)$. So $\sum_{v \in V} \deg(v) = 2m$ (the handshake lemma) and $\sum_{v \in V} \pi_v = 1$.

$\pi P = \pi$. For every vertex u, by Equation (2) and $\deg(v) \geqslant 1$, $(\pi P)_u = \sum_{v \sim u} \frac{\deg(v)}{2m} \cdot \frac{1}{\deg(v)} = \sum_{v \sim u} \frac{1}{2m} = \frac{\deg(u)}{2m} = \pi_u$, since u has $\deg(u)$ neighbours. □

For a d-regular graph on n vertices, $2m = nd$, so $\pi_v = 1/n$ is uniform. For the triangle it is $\pi = (1/3, 1/3, 1/3)$.

3 Convergence to the Stationary Distribution

Convergence of vectors is entry by entry. We state the theorem without proof.

Theorem 19 ▶ Convergence of Random Walks

Let G be a connected, non-bipartite graph with m edges. Then for every starting distribution p_0,

$$p_t = p_0 P^t \longrightarrow \pi \qquad \text{as } t \to \infty, \qquad \text{where } \pi_v = \frac{\deg(v)}{2m}.$$

So π is the only stationary distribution of the walk.

Week: 4 (September 14–18, 2026)

Lecture Date: Tuesday, September 15, 2026

Lecture Topic: Qubits and Dirac Notation, Unitary Matrices, Quantum Gates and Circuits, Tensor Products, Entanglement

This lecture covers qubits and Dirac (bra-ket) notation, unitary matrices, quantum circuits with the gates X, H and CNOT, the tensor product, and entangled states.

1 Qubits and Dirac Notation

A classical *bit* takes one of the two values 0 and 1. A quantum bit can be in a combination of both.

Definition 13 ▶ Qubit, n-Qubit State

The state of one **qubit** is a unit vector in $\mathbb{C}^2$,

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \alpha\begin{pmatrix}1\\ 0\end{pmatrix} + \beta\begin{pmatrix}0\\ 1\end{pmatrix} = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

where $|0\rangle = \begin{pmatrix}1\\ 0\end{pmatrix}$ and $|1\rangle = \begin{pmatrix}0\\ 1\end{pmatrix}$ form the **computational basis**. The complex numbers α, β are the **amplitudes**. More generally, the state of n qubits is a unit vector in $\mathbb{C}^{2^n}$,

$$|\phi\rangle = \sum_{x\in\{0,1\}^n} \alpha_x|x\rangle, \qquad \sum_{x\in\{0,1\}^n} |\alpha_x|^2 = 1,$$

where $|x\rangle$ is the standard basis vector with a 1 in the position indexed by the bit string x (the 2^n strings are listed in increasing binary order) and 0 elsewhere. Measuring the state gives the outcome x with probability $|\alpha_x|^2$.

Two qubits. The state of two qubits has four amplitudes: $(\alpha, \beta, \gamma, \delta)^T = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$ with $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$, where $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ are the four standard basis column vectors of $\mathbb{C}^4$, in this order.

Exercise 9. What is the third entry of the column vector $|00\rangle$?

Solution. It is 0. The third entry is the amplitude of $|10\rangle$, and $|00\rangle = 1\cdot|00\rangle + 0\cdot|01\rangle + 0\cdot|10\rangle + 0\cdot|11\rangle$, so $|00\rangle = (1,0,0,0)^T$. □

Writing a column vector as a combination of the kets $|x\rangle$ is *Dirac notation*. The state $\frac{1}{\sqrt{2}}|0000\rangle + \frac{1}{\sqrt{2}}|1111\rangle$ of four qubits is a column vector with $2^4 = 16$ entries: $1/\sqrt{2}$ in the first and in the last position and 0 in the other fourteen.

1.1 Kets, Bras and the Dagger

A *ket* $|\phi\rangle$ is a column vector. The corresponding *bra* is the row vector $\langle\phi| = (|\phi\rangle)^\dagger$, where † ("dagger") is the *conjugate transpose*: transpose and replace every entry by its complex conjugate.

For $|\phi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$ we get $\langle\phi| = \begin{pmatrix}\overline{\alpha} & \overline{\beta}\end{pmatrix}$. The dagger equals the inverse only for unitary matrices (Definition 14).

A bra times a ket is a number, the *inner product*, and it gives the (Euclidean) norm. For $|x\rangle, |y\rangle \in \mathbb{C}^n$ with entries x_i, y_i,

$$\langle x|y\rangle = \langle x||y\rangle = \sum_{i=1}^{n} \overline{x_i}\, y_i, \qquad \||y\rangle\|^2 = \langle y|y\rangle = \sum_{i=1}^{n} |y_i|^2.$$

The inner product is linear in the second argument and conjugate-linear in the first, and $\langle y|x\rangle = \overline{\langle x|y\rangle}$. We use two rules for the dagger: $(A^\dagger)^\dagger = A$, and $(AB)^\dagger = B^\dagger A^\dagger$ for matrices A, B whose product is defined. So $(U|x\rangle)^\dagger = \langle x|U^\dagger$.

2 Unitary Matrices

Definition 14 ▶ Unitary Matrix

An $n \times n$ complex matrix U is **unitary** if $U^\dagger U = I$.

Exercise 10. Let U be an $n \times n$ complex matrix. Show that $\|U|x\rangle\| = \||x\rangle\|$ for all $|x\rangle \in \mathbb{C}^n$ if and only if U is unitary ($U^\dagger U = I$).

Proof. Unitary implies norm preserving. If $U^\dagger U = I$, then for every $|x\rangle$, $\|U|x\rangle\|^2 = (U|x\rangle)^\dagger U|x\rangle = \langle x|U^\dagger U|x\rangle = \langle x|x\rangle = \||x\rangle\|^2$. Norms are non-negative, so $\|U|x\rangle\| = \||x\rangle\|$.
Norm preserving implies unitary. Assume $\|U|x\rangle\| = \||x\rangle\|$ for all $|x\rangle$, and let $M = U^\dagger U - I$.

1. M is Hermitian: $M^\dagger = (U^\dagger U)^\dagger - I^\dagger = U^\dagger (U^\dagger)^\dagger - I = M$.
2. Every matrix U has $\|U|x\rangle\|^2 = \langle x|U^\dagger U|x\rangle$. So for every $|x\rangle$, $\langle x|M|x\rangle = \langle x|U^\dagger U|x\rangle - \langle x|x\rangle = \|U|x\rangle\|^2 - \||x\rangle\|^2 = 0$.

Fix $|x\rangle, |y\rangle \in \mathbb{C}^n$ and let $z = \langle x|M|y\rangle$. A number is its own transpose, so its conjugate equals its dagger. By property 1, $\overline{z} = (\langle x|M|y\rangle)^\dagger = \langle y|M^\dagger|x\rangle = \langle y|M|x\rangle$. Apply property 2 to $|x\rangle + |y\rangle$, whose bra is $\langle x| + \langle y|$, and to $|x\rangle + i|y\rangle$, whose bra is $\langle x| - i\langle y|$ because the dagger conjugates. The terms $\langle x|M|x\rangle$ and $\langle y|M|y\rangle$ are 0 by property 2, so

$$0 = (\langle x| + \langle y|)M(|x\rangle + |y\rangle) = \langle x|M|y\rangle + \langle y|M|x\rangle = z + \overline{z} = 2\,\mathrm{Re}\, z,$$
$$0 = (\langle x| - i\langle y|)M(|x\rangle + i|y\rangle) = i\langle x|M|y\rangle - i\langle y|M|x\rangle = i(z - \overline{z}) = -2\,\mathrm{Im}\, z.$$

So $\langle x|M|y\rangle = z = 0$ for all $|x\rangle, |y\rangle$. Take $|x\rangle = e_j$ and $|y\rangle = e_k$, the jth and kth standard basis vectors. Then Me_k is the kth column of M and $e_j^\dagger$ picks out its jth entry, so $0 = e_j^\dagger M e_k = M_{jk}$ for all j, k. Then $M = 0$, which is $U^\dagger U = I$. □

Remark. For a square matrix, $U^\dagger U = I$ also gives $UU^\dagger = I$. If $U|x\rangle = 0$ then $|x\rangle = U^\dagger U|x\rangle = 0$, so U is injective and so invertible. Multiplying $U^\dagger U = I$ by U^{-1} on the right gives $U^\dagger = U^{-1}$, so $UU^\dagger = I$. So every unitary matrix is invertible, and its inverse $U^\dagger$ is again unitary.

A quantum gate is a linear map that sends every state (every unit vector) to a state. Every vector is a multiple of a unit vector, so such a map preserves the norm of all vectors. By Exercise 10 it is unitary. Conversely, every unitary matrix maps states to states.

3 Quantum Circuits and One-Qubit Gates

A *quantum circuit* is drawn with one horizontal wire per qubit. Time runs from left to right, and a box on a wire is a gate (a unitary matrix) applied to that qubit (Figure 9).

Columns of a matrix. Let A be an $m \times n$ matrix with columns $a_1, \ldots, a_n$, and let $x = (x_1, \ldots, x_n)^T$. Then $Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n$. So $Ae_j = a_j$: *the jth column of a gate's matrix is the image of the jth basis state.*

The X gate. The *Pauli X gate* is the quantum NOT gate, with $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$. By linearity,

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad X(\alpha|0\rangle + \beta|1\rangle) = \alpha|1\rangle + \beta|0\rangle.$$

X is real and symmetric, so $X^\dagger = X$ and $X^\dagger X = X^2 = I$. So X is unitary.

Definition 15 ▶ Hadamard Gate

The **Hadamard gate** H is the one-qubit gate with

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \qquad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{3}$$

Both $H|0\rangle$ and $H|1\rangle$ are equal superpositions: measuring gives 0 and 1 with probability $1/2$ each.

Lemma 20 ▶ Hadamard Twice Is the Identity

$H^2 = I$. So H is unitary, and

$$H\,\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |0\rangle, \qquad H\,\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |1\rangle.$$

Proof. The diagonal entries of H^2 are $\frac{1}{2}(1 \cdot 1 + 1 \cdot 1) = 1$ and $\frac{1}{2}(1 \cdot 1 + (-1)(-1)) = 1$, and both off-diagonal entries are $\frac{1}{2}(1 \cdot 1 + 1 \cdot (-1)) = 0$, so $H^2 = I$. H is real and symmetric, so $H^\dagger = H$ and $H^\dagger H = H^2 = I$. By Equation (3), the two states in the statement are $H|0\rangle$ and $H|1\rangle$, and $HH|0\rangle = |0\rangle$, $HH|1\rangle = |1\rangle$. □

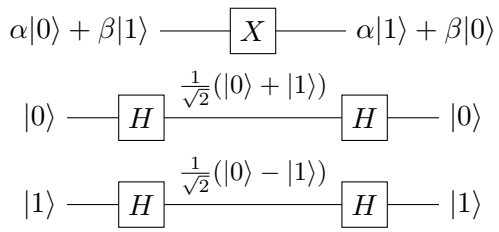


Figure 9: Top: the X gate. Below: the Hadamard gate applied twice (Lemma 20).

4 The CNOT Gate

Definition 16 ▶ CNOT Gate

The **controlled-NOT gate** (CNOT) is the two-qubit gate that maps the basis states by

$$|a\rangle|b\rangle \longmapsto |a\rangle|b \oplus a\rangle, \qquad a, b \in \{0, 1\},$$

where $\oplus$ is XOR (addition modulo 2). The first qubit $|a\rangle$ is the **control** qubit and is left unchanged. The second qubit $|b\rangle$ is the **target** qubit and is flipped exactly when $a = 1$. See Figure 10.

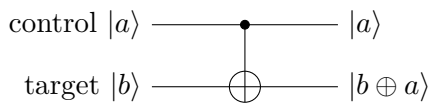


Figure 10: The CNOT gate: a filled dot on the control qubit and $\oplus$ on the target qubit.

Lemma 21 ▶ Matrix of the CNOT Gate

The matrix of the CNOT gate, with rows and columns indexed by $00, 01, 10, 11$, is the permutation matrix

$$U_{\text{CNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

It is unitary.

Proof. The columns of the matrix of a gate are the images of the basis states. By Definition 16, the images of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ are $|00\rangle, |01\rangle, |11\rangle, |10\rangle$, which are $(1,0,0,0)^T$, $(0,1,0,0)^T$, $(0,0,0,1)^T$, $(0,0,1,0)^T$. These are the four columns of U_{CNOT}, in this order.

The map $\pi\colon (a,b) \mapsto (a, b \oplus a)$ on $\{0,1\}^2$ is its own inverse, since $\pi(\pi(a,b)) = (a, b \oplus a \oplus a) = (a,b)$. So π is a permutation of the four basis states, and U_{CNOT} is a *permutation matrix*: every column is a standard basis vector, and different columns are different basis vectors.

Every permutation matrix P is unitary. The (j,k) entry of P^TP is the inner product of the jth and the kth column of P. These are standard basis vectors, equal if $j = k$ and different if $j \neq k$. So $P^TP = I$, and since P is real, $P^\dagger P = P^TP = I$. □

In block form, $U_{\text{CNOT}} = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}$. If the control is 0 nothing happens, and if it is 1 the gate X is applied to the target.

4.1 Classical Gates Are Not Reversible

By Exercise 10 and the remark after it, every quantum gate is unitary, so applying $U^\dagger$ undoes it. Classical gates that take two bits to one bit, such as AND, OR and XOR, are *not bijective*, so *not reversible*, and so *not unitary*.

Lemma 22 ▶ XOR Is Not Reversible

The classical gate XOR: $\{0,1\}^2 \to \{0,1\}$, $(a,b) \mapsto a \oplus b$, is not injective, so its input cannot be recovered from its output. The linear map $T\colon \mathbb{C}^4 \to \mathbb{C}^2$ with $T|ab\rangle = |a \oplus b\rangle$ does not preserve norms, so it is not unitary. The same holds for every gate with two input bits and one output bit.

Proof. $\mathrm{XOR}(0,0) = 0 = \mathrm{XOR}(1,1)$, so two different inputs have the same output. For the linear map, $T(|00\rangle - |11\rangle) = |0\rangle - |0\rangle = 0$, although $\||00\rangle - |11\rangle\| = \sqrt{2} \neq 0$. A general gate $g\colon \{0,1\}^2 \to \{0,1\}$ has four inputs and only two outputs, so by the pigeonhole principle two inputs $u \neq v$ have $g(u) = g(v)$. The same argument applies with $|u\rangle - |v\rangle$ (for AND, $u = 00$ and $v = 01$, and for OR, $u = 01$ and $v = 11$). □

The CNOT gate is the reversible version of XOR. It computes $b \oplus a$ on the target wire and keeps a on the control wire, so from $(a, b \oplus a)$ we recover $b = (b \oplus a) \oplus a$.

5 Tensor Products and Entanglement

Definition 17 ▶ Tensor Product

Let $A = (a_{ij})$ be an $m \times n$ matrix and let B be a $p \times q$ matrix. The **tensor product** (Kronecker product) $A \otimes B$ is the $mp \times nq$ block matrix

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & & & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}.$$

Column vectors are matrices with one column, so

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}.$$

For example, $|1\rangle \otimes |0\rangle = (0,0,1,0)^T = |10\rangle$, and in general $|a\rangle|b\rangle = |a\rangle \otimes |b\rangle = |ab\rangle$ for $a, b \in \{0,1\}$. In terms of kets, $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$. If A is applied to the first qubit and B to the second, the two-qubit gate is $A \otimes B$, and

$$(A \otimes B)(|x\rangle \otimes |y\rangle) = A|x\rangle \otimes B|y\rangle. \tag{4}$$

If $x_1, \ldots, x_n$ are the entries of $|x\rangle$, then $|x\rangle \otimes |y\rangle$ consists of the blocks $x_1|y\rangle, \ldots, x_n|y\rangle$. So the ith block of the left-hand side is $\sum_j a_{ij}B(x_j|y\rangle) = \left(\sum_j a_{ij}x_j\right)B|y\rangle$, which is the ith block of the right-hand side.

Definition 18 ▶ Product State, Entangled State

A two-qubit state $|\psi\rangle$ is a **product state** if $|\psi\rangle = |\phi_1\rangle \otimes |\phi_2\rangle$ for some one-qubit states $|\phi_1\rangle$, $|\phi_2\rangle$. Otherwise $|\psi\rangle$ is **entangled**.

Exercise 11. Show that the *Bell state* $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$ is entangled: there are no vectors

$|\phi_1\rangle, |\phi_2\rangle \in \mathbb{C}^2$ with

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = |\phi_1\rangle \otimes |\phi_2\rangle.$$

Proof. Suppose there are, and write $|\phi_1\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$ and $|\phi_2\rangle = \begin{pmatrix}\gamma\\ \delta\end{pmatrix}$. By Definition 17, comparing the four entries of the two sides gives $\alpha\gamma = 1/\sqrt{2}$, $\alpha\delta = 0$, $\beta\gamma = 0$ and $\beta\delta = 1/\sqrt{2}$. From $\alpha\delta = 0$ we get $\alpha = 0$ or $\delta = 0$. If $\alpha = 0$ then $\alpha\gamma = 0 \neq 1/\sqrt{2}$, and if $\delta = 0$ then $\beta\delta = 0 \neq 1/\sqrt{2}$. Both cases are contradictions. □

Week: 4 (September 14–18, 2026)

Lecture Date: Thursday, September 17, 2026

Lecture Topic: Linear Algebra Review: Column Space, Rank, Linear Independence, Orthogonality

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In this lecture, we review linear algebra: the product Ax as a combination of the columns of A, linear independence, column rank and row rank, and orthogonality.

1 The Product Ax as a Combination of Columns

Let $A \in \mathbb{R}^{3\times 3}$ be the matrix with columns $v_1 = (2,3,5)^T$, $v_2 = (1,1,7)^T$, $v_3 = (3,4,12)^T$. Then

$$A = \begin{bmatrix}2 & 1 & 3\\ 3 & 1 & 4\\ 5 & 7 & 12\end{bmatrix}, \qquad Ax = \begin{bmatrix}2x_1 + x_2 + 3x_3\\ 3x_1 + x_2 + 4x_3\\ 5x_1 + 7x_2 + 12x_3\end{bmatrix} = x_1v_1 + x_2v_2 + x_3v_3. \tag{5}$$

Here $v_3 = v_1 + v_2$, so $Ax = (x_1 + x_3)\,v_1 + (x_2 + x_3)\,v_2$.

In general, an $m \times n$ matrix A with columns $a_1, \dots, a_n$ defines the linear map $T\colon \mathbb{R}^n \to \mathbb{R}^m$, $T(x) = Ax = x_1a_1 + \cdots + x_na_n$. Its *image* is the *column space* of A:

$$\mathrm{Im}(A) = \{Ax \mid x \in \mathbb{R}^n\} = \mathrm{span}(a_1, \dots, a_n).$$

2 Linear Independence and Rank

Definition 19 ▶ Linear Independence

Vectors $v_1, \dots, v_k$ are **linearly independent** if

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0 \quad \Longrightarrow \quad c_1 = c_2 = \cdots = c_k = 0.$$

Otherwise they are **linearly dependent**. If $v_1, \dots, v_k$ are linearly independent, we say that a vector v is *independent of* them if $v, v_1, \dots, v_k$ are linearly independent.

Definition 20 ▶ Column Rank and Row Rank

The **column rank** $\mathrm{rank}_c(A)$ of a matrix A is the maximum number of linearly independent columns of A. The **row rank** $\mathrm{rank}_r(A)$ is the maximum number of linearly independent rows of A.

Lemma 23 ▶ More Vectors Than a Spanning Set

Let $W = \operatorname{span}(z_1, \ldots, z_r)$. Then any $r+1$ vectors $y_1, \ldots, y_{r+1}$ of W are linearly dependent.

Proof. A homogeneous system of r linear equations in more than r unknowns has a nonzero solution, by induction on r ($r = 0$: any nonzero vector). If no equation contains c_1, set $c_1 = 1$ and the other unknowns to 0. Otherwise solve one equation for c_1 and substitute into the others: the remaining $r-1$ equations in more than $r-1$ unknowns have a nonzero solution by induction, and the solved equation gives c_1.

Now write $y_j = \sum_{l=1}^{r} t_{lj} z_l$ and take a nonzero solution c of the r equations $\sum_{j=1}^{r+1} t_{lj} c_j = 0$ ($1 \leqslant l \leqslant r$). Then $\sum_j c_j y_j = \sum_l \big(\sum_j t_{lj} c_j\big) z_l = 0$. □

So all bases of W have the same size (a larger one would have more vectors than a set spanning W), the *dimension* $\dim(W)$.

Lemma 24 ▶ Column Rank Is the Dimension of the Image

Let A be an $m \times n$ matrix with $r = \operatorname{rank}_c(A) \geqslant 1$. Then any r linearly independent columns of A form a basis of $\operatorname{Im}(A)$. So

$$\operatorname{rank}_c(A) = \dim(\operatorname{Im}(A)).$$

Proof. Let $w_1, \ldots, w_r$ be independent columns and a any column of A. By the definition of r, the vectors $a, w_1, \ldots, w_r$ are dependent (trivially so if a is one of the w_l): $\beta a + \beta_1 w_1 + \cdots + \beta_r w_r = 0$ with coefficients not all zero. Here $\beta \neq 0$, since otherwise this is a nontrivial relation among the w_l, so $a = -\frac{1}{\beta}(\beta_1 w_1 + \cdots + \beta_r w_r)$. So every column, and hence every combination of columns, lies in $\operatorname{span}(w_1, \ldots, w_r)$: the independent w_l span $\operatorname{Im}(A)$, so they form a basis of it and $\operatorname{rank}_c(A) = r = \dim(\operatorname{Im}(A))$. □

Exercise 12. For the matrix A of Equation (5), find $\operatorname{rank}_c(A)$, $\operatorname{Im}(A)$ and $\operatorname{rank}_r(A)$, and express every row of A as a combination of [1 0 1] and [0 1 1].

Solution. *Columns.* v_1, v_2 are independent: $av_1 + bv_2 = 0$ gives $2a + b = 0$ and $3a + b = 0$, so $a = 0$ and then $b = 0$. The three columns are dependent, since $v_1 + v_2 - v_3 = 0$. So $\operatorname{rank}_c(A) = 2$, and by Lemma 24, $\operatorname{Im}(A) = \operatorname{span}(v_1, v_2)$, of dimension 2.
Rows. Let $u_1 = [1\ 0\ 1]$ and $u_2 = [0\ 1\ 1]$, so that $au_1 + bu_2 = [a \quad b \quad a+b]$. Then

$$[2\ 1\ 3] = 2u_1 + 1u_2, \qquad [3\ 1\ 4] = 3u_1 + 1u_2, \qquad [5\ 7\ 12] = 5u_1 + 7u_2.$$

The three rows are dependent, since $16\,[2\ 1\ 3] - 9\,[3\ 1\ 4] - [5\ 7\ 12] = 0$, so $\operatorname{rank}_r(A) \leqslant 2$. The first two rows are independent: $a\,[2\ 1\ 3] + b\,[3\ 1\ 4] = 0$ gives $a + b = 0$ (second entry) and $2a + 3b = 0$ (first entry), so $a = b = 0$. So $\operatorname{rank}_r(A) = 2$. □

Exercise 13. Show that $\operatorname{rank}_r(A) = \operatorname{rank}_c(A)$ for every $m \times n$ matrix A. This common value is called the **rank** of A.

Proof. $\operatorname{rank}_r(A) \leqslant \operatorname{rank}_c(A)$. Let $r = \operatorname{rank}_c(A)$. If $r = 0$, every column of A is zero (a single nonzero vector is independent), so $A = 0$ and $\operatorname{rank}_r(A) = 0$. Let $r \geqslant 1$, let $w_1, \ldots, w_r$ be independent columns of A, and write w_{il} for the ith entry of w_l. By Lemma 24, every column a_j of A is $a_j = \sum_{l=1}^{r} s_{lj}\, w_l$ for some numbers s_{lj}. Let $\rho_1, \ldots, \rho_r$ be the rows of the $r \times n$ matrix $S = (s_{lj})$. Entrywise, $a_{ij} = \sum_{l=1}^{r} w_{il}\, s_{lj}$; reading this for fixed i and

$j = 1, \dots, n$,

$$\text{row } i \text{ of } A \;=\; \sum_{l=1}^{r} w_{il}\, \rho_l.$$

So all rows of A lie in $\operatorname{span}(\rho_1, \dots, \rho_r)$, in which any $r+1$ vectors are dependent (Lemma 23): $\operatorname{rank}_r(A) \leqslant r = \operatorname{rank}_c(A)$.
Equality. Applying this to the transpose A^T, whose rows are the columns of A and whose columns are the rows of A, $\operatorname{rank}_c(A) = \operatorname{rank}_r(A^T) \leqslant \operatorname{rank}_c(A^T) = \operatorname{rank}_r(A)$. Together with the first part, equality holds. □

3 Orthogonality

The results of this section hold over $\mathbb{R}$ and over $\mathbb{C}$. The inner product is $\langle a, b\rangle = a \cdot b = \sum_{i=1}^{n} a_i b_i$ for $a, b \in \mathbb{R}^n$, and $\langle a, b\rangle = \sum_{i=1}^{n} \overline{a_i}\, b_i$ (linear in b) for $a, b \in \mathbb{C}^n$. In both cases $\langle a, a\rangle = \sum_{i=1}^{n} |a_i|^2 = \|a\|^2$, which is 0 only for $a = 0$.

Definition 21 ▶ Orthogonality

Two vectors a and b are **orthogonal** if $\langle a, b\rangle = 0$.

Exercise 14. Let $v_1, \dots, v_k$ be linearly independent vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$, and let $v \neq 0$ be orthogonal to each of them. Show that $v, v_1, \dots, v_k$ are linearly independent, so $v \notin \operatorname{span}(v_1, \dots, v_k)$.

Proof. Suppose that $c\,v + c_1 v_1 + \cdots + c_k v_k = 0$. Take the inner product of v with both sides (it is linear in the second argument) and use $\langle v, v_i\rangle = 0$:

$$0 = c\,\langle v, v\rangle + \sum_{i=1}^{k} c_i\,\langle v, v_i\rangle = c\,\|v\|^2.$$

Since $v \neq 0$, $\|v\|^2 > 0$, so $c = 0$. Then $c_1 v_1 + \cdots + c_k v_k = 0$, and the independence of $v_1, \dots, v_k$ gives $c_1 = \cdots = c_k = 0$. If $v = \sum_i \alpha_i v_i$, then $1 \cdot v - \sum_i \alpha_i v_i = 0$ would be a combination with a nonzero coefficient that gives zero, so $v \notin \operatorname{span}(v_1, \dots, v_k)$. □

Remark. The assumption $v \neq 0$ is needed. The zero vector is orthogonal to every vector, but no set containing it is independent ($1 \cdot 0 = 0$ has a nonzero coefficient).

Week: 5 (September 21–25, 2026)

Lecture Date: Thursday, September 24, 2026

Lecture Topic: Deutsch's Algorithm, Phase Flip, Walsh–Hadamard Transform, Deutsch–Jozsa Algorithm

Deutsch's algorithm and the Deutsch–Jozsa algorithm beat every classical algorithm in the query model. The second uses the *phase flip* form of a query and the n-qubit Hadamard transform.

1 Deutsch's Algorithm

Problem 2 ▶ Deutsch's Problem

Given a function $f\colon \{0,1\} \to \{0,1\}$ as a black box, determine whether f is *constant* ($f(0) = f(1)$) or *balanced* ($f(0) \neq f(1)$).

Classically we need 2 queries: after one query, say $f(0)$, both answers are still possible.

1.1 The Quantum Query U_f

A quantum circuit is reversible, but f need not be. So we use the reversible *quantum version* U_f of f, a controlled-f gate like the CNOT gate of Definition 16 (Figure 11):

$$U_f\colon |a\rangle|b\rangle \longmapsto |a\rangle|b \oplus f(a)\rangle.$$

Applying U_f twice gives back $|a\rangle|b\rangle$, so U_f is reversible. One application of U_f counts as one query to f.

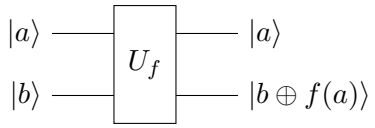


Figure 11: The quantum query U_f.

1.2 The Circuit

Deutsch's algorithm (Figure 12) applies a Hadamard gate to each of $|0\rangle$ and $|1\rangle$, then U_f, then a Hadamard gate to each qubit, and measures.

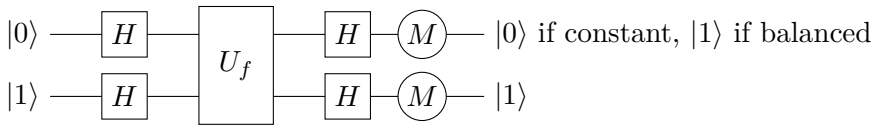


Figure 12: Deutsch's algorithm.

By Equation (3), for $a \in \{0,1\}$,

$$H|a\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle) = \frac{1}{\sqrt{2}} \sum_{b\in\{0,1\}} (-1)^{ab}|b\rangle. \tag{6}$$

Exercise 15. Compute $H\,\frac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle)$ for $a \in \{0,1\}$.

Solution. By linearity and Equation (6) for $H|0\rangle$ and $H|1\rangle$,

$$H\,\tfrac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle) = \tfrac{1}{2}(|0\rangle + |1\rangle) + \tfrac{(-1)^a}{2}(|0\rangle - |1\rangle) = \tfrac{1+(-1)^a}{2}|0\rangle + \tfrac{1-(-1)^a}{2}|1\rangle.$$

For $a = 0$ this is $|0\rangle$ and for $a = 1$ it is $|1\rangle$, so the result is $|a\rangle$. This agrees with $H^2 = I$ (Lemma 20). □

Exercise 16. In the circuit of Figure 12 the second qubit is measured as $|1\rangle$. What does the measurement of the first qubit give? Show that the circuit makes one query to f and that measuring its first qubit gives $f(0) \oplus f(1)$ with probability 1. So the outcome is 0 if f is constant and 1 if f is balanced.

Proof. The circuit contains U_f once, so it makes one query. After the first two Hadamard gates the state is (by Equation (4)) $|\psi_1\rangle = H|0\rangle \otimes H|1\rangle = \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)$. For $c \in \{0,1\}$ we have $|0 \oplus c\rangle - |1 \oplus c\rangle = (-1)^c(|0\rangle - |1\rangle)$ (for $c = 1$ the two kets are swapped). So U_f maps $|a\rangle(|0\rangle - |1\rangle)$ to $(-1)^{f(a)}|a\rangle(|0\rangle - |1\rangle)$. By linearity,

$$|\psi_2\rangle = U_f|\psi_1\rangle = \frac{1}{\sqrt{2}}\Big((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\Big) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Since $(-1)^{f(1)} = (-1)^{f(0)}(-1)^{f(0)\oplus f(1)}$, the first qubit is $(-1)^{f(0)}\,\frac{1}{\sqrt{2}}\Big(|0\rangle + (-1)^{f(0)\oplus f(1)}|1\rangle\Big)$. Up to the sign $(-1)^{f(0)}$, this is the state of Exercise 15 with $a = f(0) \oplus f(1)$. The second qubit is $H|1\rangle$. So the last two Hadamard gates give $|\psi_3\rangle = (-1)^{f(0)}|f(0) \oplus f(1)\rangle|1\rangle$. The global phase $(-1)^{f(0)}$ does not affect the measurement. The first qubit gives $f(0) \oplus f(1)$ with probability 1, and the second always gives 1. □

2 Phase Flip and the n-Qubit Hadamard Transform

If the second (target) qubit of U_f is in the state $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, then, by the computation in the proof of Exercise 16, for every input x (also for $x \in \{0,1\}^n$, where $U_f\colon |x\rangle|b\rangle \mapsto |x\rangle|b \oplus f(x)\rangle$),

$$U_f\,|x\rangle\tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = (-1)^{f(x)}\,|x\rangle\tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

We drop the target qubit and write the query in the *phase flip* form

$$U_f\colon |x\rangle \longmapsto (-1)^{f(x)}|x\rangle.$$

By linearity, U_f maps $\sum_x \alpha_x|x\rangle$ to $\sum_x \alpha_x(-1)^{f(x)}|x\rangle$.

2.1 The n-Qubit Hadamard Transform

For $n = 2$ let $|x\rangle = |x_1x_2\rangle = |x_1\rangle|x_2\rangle$. By Equation (4) and then Equation (6) on each qubit,

$$\begin{aligned}(H \otimes H)|x\rangle = H|x_1\rangle \otimes H|x_2\rangle &= \left(\frac{1}{\sqrt{2}}\sum_{y_1\in\{0,1\}}(-1)^{x_1y_1}|y_1\rangle\right)\left(\frac{1}{\sqrt{2}}\sum_{y_2\in\{0,1\}}(-1)^{x_2y_2}|y_2\rangle\right)\\ &= \frac{1}{2}\sum_{y\in\{0,1\}^2}(-1)^{x_1y_1+x_2y_2}|y\rangle = \frac{1}{2}\sum_{y\in\{0,1\}^2}(-1)^{x\cdot y}|y\rangle,\end{aligned}$$

where $x \cdot y = x_1y_1 \oplus x_2y_2$, since only the parity of the exponent matters. For example, $(H \otimes H)|01\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$. In general, for $x, y \in \{0,1\}^n$, the inner product of the bit strings modulo 2 is

$$x \cdot y = x_1y_1 \oplus x_2y_2 \oplus \cdots \oplus x_ny_n.$$

Lemma 25 ▶ Walsh–Hadamard Transform

The n-qubit Hadamard transform $H^{\otimes n} = H \otimes H \otimes \cdots \otimes H$ (the *Walsh–Hadamard transform*) acts on a basis state $|x\rangle = |x_1x_2\cdots x_n\rangle$ as

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle.$$

Proof. By Equation (4) applied repeatedly, $H^{\otimes n}|x\rangle = H|x_1\rangle \otimes \cdots \otimes H|x_n\rangle$. By Equation (6) the ith factor is $\frac{1}{\sqrt{2}} \sum_{y_i\in\{0,1\}}(-1)^{x_iy_i}|y_i\rangle$. Multiplying out the n sums,

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x_1y_1+\cdots+x_ny_n}|y\rangle,$$

and (-1) to the power $x_1y_1 + \cdots + x_ny_n$ only depends on its parity $x \cdot y$. □

For $x = 0^n = 00\cdots0$ we have $x \cdot y = 0$ for every y, so $H^{\otimes n}|00\cdots0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n}|x\rangle$, the uniform superposition.

3 Deutsch–Jozsa Algorithm

Problem 3 ▶ Deutsch–Jozsa Problem

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, with the *promise* that f is either constant or balanced (that is, $f(x) = 0$ for exactly half of the 2^n inputs), determine which is the case.

This is a *promise problem*. Let $N = 2^n$ be the number of inputs.

3.1 Classical Query Complexity

Deterministic: in the worst case we need $2^{n-1} + 1 = N/2 + 1$ queries, since after $N/2$ equal values f may still be either constant or balanced. The same holds for a randomized algorithm that has to succeed with probability 1.

Randomized (Monte Carlo): query f on α independent uniformly random inputs and output "balanced" if two answers differ, "constant" otherwise. If f is constant, we are always right. If f is balanced, every answer is 0 or 1 with probability $1/2$ each. We fail only if all α answers agree, which has probability $2 \cdot (1/2)^\alpha = (1/2)^{\alpha-1}$. A constant number of queries gives a constant error probability, and $\alpha = \mathcal{O}(\log N)$ queries give error probability $1/N$.

Quantum: the Deutsch–Jozsa algorithm needs 1 query (and n measurements) and makes no error.

3.2 The Algorithm

The circuit (Figure 13) is Deutsch's circuit with n input qubits in $|0\rangle$. After U_f every qubit gets a second Hadamard gate and is measured. The last qubit always gives 1.

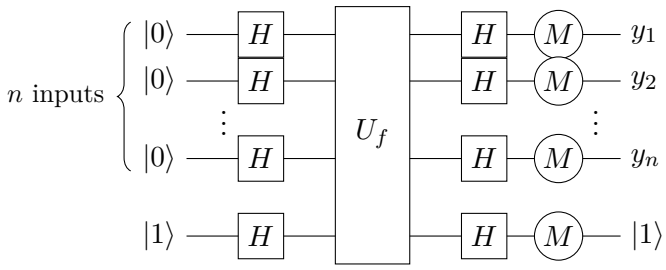


Figure 13: The Deutsch–Jozsa algorithm.

Exercise 17. In the circuit of Figure 13 the state of the n input qubits before the measurement is $\frac{1}{2^n}\sum_y\sum_x(-1)^{f(x)+x\cdot y}|y\rangle$. Show that measuring them gives $y = 0^n$ with probability 1 if f is constant, and with probability 0 if f is balanced. So if any of these n measurements gives 1, then f is balanced; otherwise f is constant.

Proof. We use the phase flip form of U_f. The last qubit is $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = H|1\rangle$ before and after the query. Its second Hadamard gate gives $|1\rangle$, so we drop it. The first layer of Hadamard gates maps the input register $|00\cdots0\rangle$ to $\frac{1}{\sqrt{2^n}}\sum_x|x\rangle$. Then the query flips the sign of every $|x\rangle$ with $f(x) = 1$. After the second layer of Hadamard gates, by Lemma 25, the state is

$$\frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}(-1)^{f(x)}H^{\otimes n}|x\rangle = \frac{1}{2^n}\sum_{y\in\{0,1\}^n}\sum_{x\in\{0,1\}^n}(-1)^{f(x)+x\cdot y}|y\rangle.$$

Since $x\cdot 0^n = 0$ for every x, the amplitude of $|0^n\rangle$ is $\frac{1}{2^n}\sum_x(-1)^{f(x)}$. If f is constant, all 2^n terms are equal and the amplitude is ±1. The amplitudes squared sum to 1, so all others are 0 and the measurement gives 0^n with probability 1. If f is balanced, half the terms are $+1$ and half are -1, so the amplitude is 0. The measurement gives 0^n with probability 0, so at least one of the n measurements gives 1. □

Week: 6 (September 28 – October 2, 2026)

Lecture Date: Tuesday, September 29, 2026

Lecture Topic: Walsh–Hadamard Transform Recap, Bernstein–Vazirani Algorithm, Simon's Algorithm

In this lecture, we see two more problems that a quantum algorithm solves with far fewer queries than any classical algorithm: the *Bernstein–Vazirani* problem and *Simon's problem*, the first exponential quantum speed-up.

Recap: quantum queries and the Walsh–Hadamard transform. Both algorithms use the query $U_f\colon |x\rangle|b\rangle \mapsto |x\rangle|b \oplus f(x)\rangle$ (Figure 11; one application is one query) and the *Walsh–Hadamard transform* (Lemma 25) $H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle$, where $x \cdot y = x_1y_1 \oplus \cdots \oplus x_ny_n$.

Lemma 26 ▶ Sum of Signs

For every $z \in \{0,1\}^n$,

$$\sum_{x\in\{0,1\}^n} (-1)^{z\cdot x} = \begin{cases} 2^n & \text{if } z = 0^n, \\ 0 & \text{if } z \neq 0^n. \end{cases}$$

Proof. If $z = 0^n$, all 2^n terms are 1. If $z \neq 0^n$, pick i with $z_i = 1$ and let e_i be the string with a single 1 in position i. The map $x \mapsto x \oplus e_i$ is its own inverse and has no fixed point, so it splits $\{0,1\}^n$ into 2^{n-1} disjoint pairs $\{x, x \oplus e_i\}$. The inner product is linear modulo 2, so $z \cdot (x \oplus e_i) = z \cdot x \oplus z_i = z \cdot x \oplus 1$, and every pair contributes $(-1)^{z\cdot x} + (-1)^{z\cdot x\oplus 1} = 0$. □

In the Deutsch–Jozsa circuit (Figure 13, Exercise 17) the query acts as the phase flip $|k\rangle \mapsto (-1)^{f(k)}|k\rangle$, and the control register after the second layer of Hadamard gates is in the state

$$\sum_{y\in\{0,1\}^n} \alpha_y|y\rangle, \qquad \alpha_y = \frac{1}{2^n} \sum_{k\in\{0,1\}^n} (-1)^{f(k)+k\cdot y}. \tag{7}$$

1 Bernstein–Vazirani Algorithm

Problem 4 ▶ Bernstein–Vazirani Problem

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, with the *promise* that

$$f(x) = s \cdot x = s_1x_1 \oplus s_2x_2 \oplus \cdots \oplus s_nx_n$$

for some unknown n-bit string s, find s.

1.1 The Algorithm

Theorem 27 ▶ Bernstein–Vazirani Algorithm

Run the Deutsch–Jozsa circuit (Figure 13) on $f(x) = s \cdot x$. Then measuring the n control qubits gives $y = s$ with probability 1. So a single query finds s.

Proof. By Equation (7) (with the summation index k renamed x), the control register is in the state $\sum_y \alpha_y |y\rangle$ with

$$\alpha_y = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x \oplus y \cdot x} = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{(s \oplus y) \cdot x},$$

because only the parity of the exponent matters and the inner product is bilinear modulo 2. By Lemma 26 with $z = s \oplus y$, $\alpha_y = 1$ if $y = s$ and $\alpha_y = 0$ if $y \neq s$. So the state is $|s\rangle$, and the measurement gives s with probability 1. □

Query complexity versus running time. The query model only counts the calls to U_f. The running time in the *quantum Turing machine* (or quantum circuit) model counts all the gates. The two are not always the same.

2 Simon's Algorithm

Problem 5 ▶ Simon's Problem

Given $f\colon \{0,1\}^n \to \{0,1\}^n$ as a black box, with the *promise* that there is a string $s \in \{0,1\}^n$ such that for all x, y,

$$f(x) = f(y) \iff x = y \text{ or } x = y \oplus s \qquad \text{(equivalently } s = x \oplus y\text{)},$$

find s.

If $s \neq 0^n$, then f is 2-*to*-1 with period s: the two inputs of a pair $\{x, x \oplus s\}$ map to the same output, and different pairs map to different outputs. If $s = 0^n$, then f is 1-*to*-1. So the problem asks whether f is 1-to-1 or 2-to-1 and, in the second case, for the period s. We write $N = 2^n$.

Query complexity. Classically, Simon's problem needs $\Omega\left(\sqrt{N}\right) = \Omega\left(\sqrt{2^n}\right)$ queries, even for randomized algorithms (next lecture). Simon's quantum algorithm uses $\mathcal{O}(n)$ queries and a Gaussian elimination on an $n \times n$ system of linear equations over $\{0,1\}$, which takes $\mathcal{O}(n^3)$ time. This is an exponential speed-up.

2.1 The Circuit

The circuit (Figure 14) has two registers of n qubits each, all starting in $|0\rangle$. Hadamard gates act on the control register before and after the single query U_f. Then the control register is measured, giving a string $y \in \{0,1\}^n$. Because f has n output bits, we use U_f in its original form $|x\rangle|b\rangle \mapsto |x\rangle|b \oplus f(x)\rangle$ and not the phase flip form.

2.2 Analysis of One Run

Theorem 28 ▶ One Run of Simon's Circuit

Let y be the string measured by the circuit of Figure 14.

1. If f is 2-to-1 with period $s \neq 0^n$, then y is uniformly distributed over the 2^{n-1} strings y with $s \cdot y = 0$: each of them has probability $1/2^{n-1}$, and every y with $s \cdot y = 1$ has probability 0.
2. If f is 1-to-1, then y is uniformly distributed over all of $\{0,1\}^n$: each string has probability $1/2^n$.

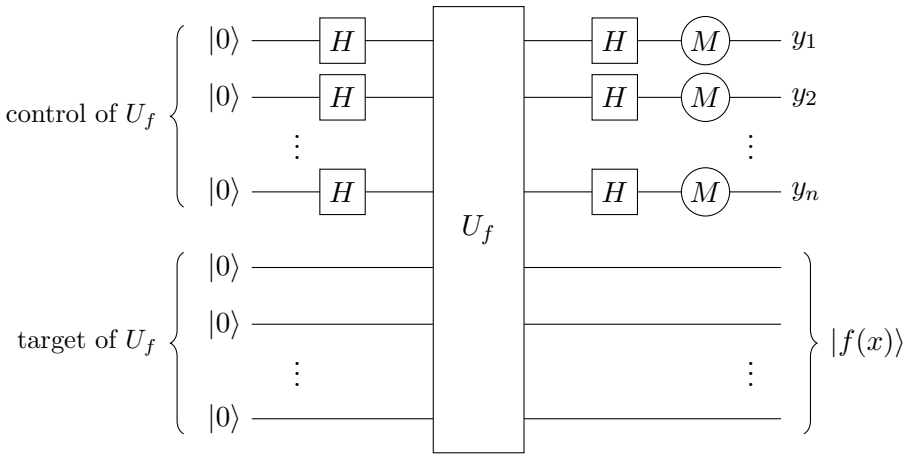


Figure 14: Simon's circuit. The target register is not measured.

Proof. By Lemma 25, the first layer of Hadamard gates gives $\frac{1}{\sqrt{2^n}}\sum_x |x\rangle|0^n\rangle$. The query U_f turns this into $\frac{1}{\sqrt{2^n}}\sum_x |x\rangle|f(x)\rangle$. The second layer of Hadamard gates on the control register (Lemma 25 applied to every $|x\rangle$) gives

$$\frac{1}{2^n}\sum_{y\in\{0,1\}^n}\sum_{x\in\{0,1\}^n}(-1)^{x\cdot y}|y\rangle|f(x)\rangle. \tag{8}$$

The amplitude $\alpha_{y,z}$ of a basis state $|y\rangle|z\rangle$ collects the x with $f(x)=z$. Measuring only the first register gives y with probability $\sum_z|\alpha_{y,z}|^2$ (the probabilities of all basis states $|y\rangle|z\rangle$ with this y add up).

Case 2: f is 1-to-1. The 2^n strings $f(x)$ are all different, so every basis state $|y\rangle|f(x)\rangle$ appears exactly once in Equation (8), with amplitude $\frac{1}{2^n}(-1)^{x\cdot y}$. So y has probability $\sum_x\left|\frac{1}{2^n}(-1)^{x\cdot y}\right|^2 = 2^n\cdot\frac{1}{2^{2n}} = \frac{1}{2^n}$.

Case 1: f is 2-to-1 with period $s\neq 0^n$. The range R of f has $|R|=2^{n-1}$ elements. For every $z\in R$ let x_z be one of its two preimages; the other is $x_z\oplus s$. Because $(x_z\oplus s)\cdot y = x_z\cdot y\oplus s\cdot y$, the amplitude of $|y\rangle|z\rangle$ in Equation (8) is

$$\frac{1}{2^n}\Big((-1)^{x_z\cdot y}+(-1)^{(x_z\oplus s)\cdot y}\Big) = \frac{1}{2^n}(-1)^{x_z\cdot y}(1+(-1)^{s\cdot y})$$

for $z\in R$ (and 0 for $z\notin R$). So y has probability

$$\sum_{z\in R}\left|\frac{1}{2^n}(1+(-1)^{s\cdot y})\right|^2 = 2^{n-1}\cdot\frac{(1+(-1)^{s\cdot y})^2}{2^{2n}} = \begin{cases}1/2^{n-1} & \text{if } s\cdot y=0,\\ 0 & \text{if } s\cdot y=1.\end{cases}$$

There are 2^{n-1} strings y with $s\cdot y=0$, because the pairing $y\mapsto y\oplus e_i$ of the proof of Lemma 26, with $s_i=1$, matches every y with $s\cdot y=0$ to a y' with $s\cdot y'=1$. □

2.3 From the Measurements to s

Each run of the circuit gives a random y with $s\cdot y=0$, which is one linear equation $s_1y_1\oplus s_2y_2\oplus\cdots\oplus s_ny_n=0$ for the unknown bits $s_1,\ldots,s_n$. Running the circuit $n-1$ times gives a homogeneous system

$$y_1\cdot s=0,\quad y_2\cdot s=0,\quad \ldots,\quad y_{n-1}\cdot s=0$$

of linear equations over the field $\{0,1\}$ (addition is $\oplus$). If $y_1,\ldots,y_{n-1}$ are *linearly independent* over $\{0,1\}$, the system has rank $n-1$. Then its solution space has dimension $n-(n-1)=1$ and

consists of 0^n and one nonzero string s'. Gaussian elimination over $\{0,1\}$ tests the independence and computes s' in $\mathcal{O}(n^3)$ time. The true s satisfies every equation. So if $s \neq 0^n$, at most $n-1$ of them can be independent, and $n-1$ independent ones are enough: if the y_i are independent and f is 2-to-1, then $s' = s$.

Algorithm 1 Simon's algorithm

run the circuit of Figure 14 $n-1$ times; let $y_1, \ldots, y_{n-1}$ be the measured strings
if $y_1, \ldots, y_{n-1}$ are linearly dependent over $\{0,1\}$: ▷ *Gaussian elimination*
    return FAIL
let s' be the unique nonzero solution of $y_1 \cdot s' = \cdots = y_{n-1} \cdot s' = 0$
if $f(s') = f(0^n)$: ▷ *two classical queries*
    return s' ▷ *f is 2-to-1 with period s'*
else:
    return 0^n ▷ *f is 1-to-1*

Theorem 29 ▶ Simon's Algorithm

Algorithm 1 makes $n+1$ queries ($n-1$ quantum, 2 classical) and $\mathcal{O}(n^3)$ other operations. It never outputs a wrong answer, and it outputs the correct s with probability at least $1/4$.

Proof. Never wrong. If the algorithm does not fail, $y_1, \ldots, y_{n-1}$ are independent and s' is the unique nonzero solution of the system. If f is 2-to-1 with period s, then s is a nonzero solution, so $s' = s$. The test $f(s') = f(0^n) = f(0^n \oplus s)$ succeeds, and the output $s' = s$ is correct. If f is 1-to-1, then $f(s') \neq f(0^n)$ because $s' \neq 0^n$, and the output $0^n = s$ is correct.

Success probability. The algorithm fails only if the y_i are dependent. The $n-1$ runs are independent (the registers are re-initialized to $|0^n\rangle|0^n\rangle$ for every run). By Theorem 28, the y_i are independent uniform samples from the 2^{n-1} strings with $s \cdot y = 0$ if f is 2-to-1, and from all of $\{0,1\}^n$ if f is 1-to-1. By Exercise 20 (next lecture) they are linearly independent with probability

$$\left(1-\frac{1}{2}\right)\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{2^3}\right)\cdots\left(1-\frac{1}{2^{n-1}}\right) \geqslant \frac{1}{4}$$

in the first case and at least $1/2$ in the second. □

2.4 A Deterministic Classical Algorithm

Exercise 18. Find a deterministic classical algorithm for Simon's problem that makes $\mathcal{O}\left(\sqrt{2^n}\right)$ queries.

Solution. We query a set S such that every nonzero string is the XOR of two elements of S. A collision in S then gives s. Let $m = \lceil n/2 \rceil$ and write every n-bit string as $x = (u, v)$ with $u \in \{0,1\}^m$ its first m bits and $v \in \{0,1\}^{n-m}$ its last $n-m$ bits. Let

$$S_1 = \{(u, 0^{n-m}) \mid u \in \{0,1\}^m\}, \qquad S_2 = \left\{(0^m, v) \mid v \in \{0,1\}^{n-m}\right\}, \qquad S = S_1 \cup S_2.$$

The algorithm. Query f on every $x \in S$. If there are $x \neq x'$ in S with $f(x) = f(x')$, output $s = x \oplus x'$. Otherwise output $s = 0^n$.
Queries. Because $n - m \leqslant m \leqslant (n+1)/2$, the algorithm makes $|S| \leqslant 2^m + 2^{n-m} \leqslant 2 \cdot 2^m \leqslant 2 \cdot 2^{(n+1)/2} = 2\sqrt{2} \cdot \sqrt{2^n} = \mathcal{O}\left(\sqrt{2^n}\right)$ queries, and it is deterministic.
Claim: every $s \neq 0^n$ is the XOR of two different elements of S. Write $s = (u, v)$. Then

$(u, 0^{n-m}) \in S_1$, $(0^m, v) \in S_2$ and $(u, 0^{n-m}) \oplus (0^m, v) = s$. The two elements are different, because otherwise $u = 0^m$ and $v = 0^{n-m}$, so $s = 0^n$.
Correctness. If f is 2-to-1 with period $s \neq 0^n$, by the claim there are $x \neq x'$ in S with $x' = x \oplus s$, so $f(x) = f(x')$ and we find a collision. Any collision $f(x) = f(x')$ with $x \neq x'$ satisfies $x' = x \oplus s$ by the promise, so the output $x \oplus x' = s$ is correct. If f is 1-to-1 there is no collision, and the output $0^n = s$ is correct. □

Week: 6 (September 28 – October 2, 2026)

Lecture Date: Thursday, October 1, 2026

Lecture Topic: Classical Query Complexity of Simon's Problem, Yao's Minmax Principle, Linear Independence in Simon's Algorithm

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In this lecture, we show that the *classical* query complexity of Simon's problem is $\Theta\left(\sqrt{N}\right)$, where $N = 2^n$, and that the strings measured by Simon's quantum algorithm are linearly independent with probability at least $1/4$.

1 Simon's Problem, Classically

In Simon's problem, $f\colon \{0,1\}^n \to \{0,1\}^n$ is given as a black box with the promise

$$f(x) = f(y) \iff x = y \quad \text{or} \quad x \oplus y = s,$$

and we have to find s (f is 1-to-1 if $s = 0^n$ and 2-to-1 with period s otherwise). Let $N = 2^n$. A classical algorithm learns f only through queries $f(q_1), f(q_2), \ldots$, and we may assume that it never queries the same input twice. A *collision* is a pair of queries $q_i \neq q_j$ with $f(q_i) = f(q_j)$. By the promise a collision gives $s = q_i \oplus q_j$.

1.1 Deterministic Algorithms

Exercise 18 gives a deterministic algorithm with $\mathcal{O}\left(\sqrt{N}\right)$ queries. This is optimal by Corollary 34.

1.2 A Randomized Algorithm

We query random inputs and wait for a collision.

Algorithm 2 Randomized classical algorithm for Simon's problem

choose k distinct inputs $q_1, \ldots, q_k$ uniformly at random (without replacement) and query them
if $f(q_i) = f(q_j)$ for some $i \neq j$:
    return $q_i \oplus q_j$
else:
    return 0^n

Theorem 30 ▶ Randomized Upper Bound

If f is 1-to-1, Algorithm 2 is always correct. If f is 2-to-1 with period s, it is wrong with probability at most $e^{-k(k-1)/(2N)}$. So with $k \geqslant \sqrt{2N \ln N} + 1$ queries (an integer at most N once $n \geqslant 3$) the algorithm is correct with probability at least $1 - 1/N$. With $k \geqslant \sqrt{2N} + 1$ queries, which is $\mathcal{O}\left(\sqrt{N}\right)$, this probability is at least $1 - 1/e$.

Proof. If f is 1-to-1 there is no collision, and the output 0^n is correct. Let f be 2-to-1 with period $s \neq 0^n$. The algorithm is wrong only if no collision occurs. Call $q_j \oplus s$ the *mirror* of q_j. A collision occurs at step i exactly when q_i is the mirror of an earlier query.

Suppose that no collision occurred among $q_1, \ldots, q_{i-1}$. Their $i-1$ mirrors are distinct, and none is among $q_1, \ldots, q_{i-1}$, because that would be a collision. Then all of them are among the $N - (i-1)$ inputs not yet queried, and q_i is uniform among those. So the probability of a collision at step i, given none before, is $(i-1)/(N-i+1) \geqslant (i-1)/N$. Multiplying over the k steps and using $1 - x \leqslant e^{-x}$,

$$\begin{aligned}\mathbb{P}(\text{no collision among } q_1, \ldots, q_k) &\leqslant \prod_{i=1}^{k}\left(1 - \frac{i-1}{N}\right) \leqslant e^{-\sum_{i=1}^{k}(i-1)/N} \\ &= e^{-k(k-1)/(2N)} \leqslant e^{-(k-1)^2/(2N)}.\end{aligned}$$

With $k - 1 \geqslant \sqrt{2N \ln N}$ this is at most $e^{-\ln N} = 1/N$, and with $k - 1 \geqslant \sqrt{2N}$ it is at most e^{-1}. □

1.3 Lower Bound for Randomized Algorithms via Yao's Principle

Theorem 31 ▶ Randomized Lower Bound

Let $n \geqslant 2$. Every randomized classical algorithm that solves Simon's problem with success probability at least $2/3$ on every input makes more than $\sqrt{N}/2$ queries on some input. So every such algorithm needs $\Omega\left(\sqrt{N}\right) = \Omega\left(\sqrt{2^n}\right)$ queries.

Lemma 32 ▶ Yao's Minmax Principle, Easy Direction

Let μ be a probability distribution on the inputs. If every deterministic algorithm that makes at most k queries errs with probability greater than δ on an input drawn from μ, then every randomized algorithm that makes at most k queries errs with probability greater than δ on some input.

Proof. A randomized algorithm R is a deterministic algorithm A_r with at most k queries for every fixed random string r. Let $\mathrm{err}(A, x) \in \{0, 1\}$ indicate that A is wrong on input x. Then

$$\max_x \mathbb{P}_r(\mathrm{err}(A_r, x) = 1) \geqslant \mathbb{E}_{x\sim\mu}[\mathbb{P}_r(\mathrm{err}(A_r, x) = 1)] = \mathbb{E}_r[\mathbb{P}_{x\sim\mu}(\mathrm{err}(A_r, x) = 1)] > \delta,$$

since a maximum is at least an average, the two expectations can be exchanged, and every inner probability on the right is greater than δ by assumption. □

The input distribution. Let $\mathcal{D}$ be the following distribution of 2-to-1 functions. Choose s uniformly among the $N-1$ nonzero strings. This splits $\{0,1\}^n$ into $N/2$ pairs $\{x, x \oplus s\}$. Assign distinct values to the pairs, uniformly at random among all such assignments.

Principle of deferred decisions. We generate the input lazily. Choose s. When the algorithm queries q_i: if $q_i = q_j \oplus s$ for an earlier query q_j (a collision), answer $f(q_j)$. Otherwise answer a uniformly random value among those not used so far ($N - (i-1)$ of them as long as no collision has occurred). This produces $\mathcal{D}$, because a uniformly random injective assignment can be built one value at a time, each uniform among the unused ones.

Lemma 33 ▶ Few Queries Rarely Collide

Let $n \geqslant 2$ and let A be a deterministic algorithm that makes $k \leqslant \sqrt{N}/2$ queries, all different, on an input drawn from $\mathcal{D}$.

1. The probability that two of the queries form a pair $\{x, x \oplus s\}$ (a collision) is at most
$$\sum_{i=1}^{k} \frac{i-1}{N - 1 - \binom{i-1}{2}} \leqslant \frac{k(k-1)}{N}.$$
2. Given that there is no collision among the k queries, and given the k answers, s is uniformly distributed over a set of at least $N/2$ strings.

Proof. Let Q_i be the event that there is no collision among the first i queries, and C_i the event $q_i \oplus s \in \{q_1, \ldots, q_{i-1}\}$ that the ith query collides with an earlier one.

What the algorithm knows about s. Fix $i \leqslant k+1$ and condition on Q_{i-1} and on the (distinct) answers $f(q_1), \ldots, f(q_{i-1})$. Since A is deterministic, the queries $q_1, \ldots, q_{i-1}$ are the same for every s that leads to these answers. By the lazy generation, this sequence of answers has probability $\prod_{j=1}^{i-1} 1/(N-j+1)$, the same for every s not of the form $q_j \oplus q_l$ with $j < l \leqslant i-1$. It is 0 for an s of this form (q_j and q_l would have collided). By Bayes' rule, given the answers, s is uniform over the nonzero strings that are not among the at most $\binom{i-1}{2}$ strings $q_j \oplus q_l$. This is a set of at least $N - 1 - \binom{i-1}{2} \geqslant N - 1 - k^2/2 \geqslant N - 1 - N/8 \geqslant N/2$ candidates (as $i - 1 \leqslant k \leqslant \sqrt{N}/2$ and $N \geqslant 4$). For $i = k+1$ this is part 2.

Part 1. If C_i is the first collision then Q_{i-1} holds, so by the union bound $\mathbb{P}(\text{collision}) \leqslant \sum_{i=1}^{k} \mathbb{P}(C_i \cap Q_{i-1})$. Given Q_{i-1} and the answers, q_i is determined, and C_i occurs exactly when s is one of the $i-1$ strings $q_i \oplus q_1, \ldots, q_i \oplus q_{i-1}$. Averaging over the answers,

$$\mathbb{P}(C_i \cap Q_{i-1}) \leqslant \frac{i-1}{N - 1 - \binom{i-1}{2}} \leqslant \frac{2(i-1)}{N}, \qquad \text{so} \qquad \mathbb{P}(\text{collision}) \leqslant \sum_{i=1}^{k} \frac{2(i-1)}{N} = \frac{k(k-1)}{N}.$$

□

Proof of Theorem 31. Let A be a deterministic algorithm with at most $\sqrt{N}/2$ queries. We may assume that it never repeats a query and makes the same number $k \leqslant \sqrt{N}/2$ of queries on every input, because extra queries whose answers are ignored do not change the output. Run it on an input drawn from $\mathcal{D}$. It is correct only if its output is s. Suppose there is no collision. Given the answers, the output is one fixed string, and s is uniform over at least $N/2$ strings by part 2 of Lemma 33. So A is correct with probability at most $2/N$. With part 1,

$$\mathbb{P}(A \text{ correct}) \leqslant \mathbb{P}(\text{collision}) + \frac{2}{N} \leqslant \frac{k(k-1)+2}{N}.$$

If $k \geqslant 2$, then $k(k-1)+2 \leqslant k^2 \leqslant N/4$ and the bound is at most $1/4$. If $k \leqslant 1$, there is no collision and the bound is $2/N \leqslant 1/2$. So A errs with probability at least $1/2 > 1/3$ on $\mathcal{D}$. By Lemma 32 with $\mu = \mathcal{D}$ and $\delta = 1/3$, every randomized algorithm with at most $\sqrt{N}/2$ queries errs with probability greater than $1/3$ on some input. So an algorithm with success probability at least $2/3$ on every input makes more than $\sqrt{N}/2$ queries on some input. □

Corollary 34 ▶ Classical Query Complexity of Simon's Problem

The classical query complexity of Simon's problem is $\Theta\left(\sqrt{N}\right) = \Theta\left(\sqrt{2^n}\right)$, both for deterministic algorithms and for randomized algorithms with constant success probability. So Simon's quantum algorithm, with $\mathcal{O}(n)$ queries, is exponentially faster than every classical algorithm.

Proof. A deterministic algorithm that is always correct is a randomized algorithm with success probability $1 \geqslant 2/3$, so Theorem 31 gives the lower bound for both kinds. The upper bounds are Exercise 18 and Theorem 30 (the bound $e^{-(k-1)^2/(2N)}$ in its proof is at most $1/3$ for $k \geqslant \sqrt{2N \ln 3} + 1$). □

Corollary 35 ▶ Decision Version

Let $n \geqslant 2$. Every randomized classical algorithm that decides, with success probability at least $2/3$ on every input, whether f is 1-to-1 or 2-to-1 makes more than $\sqrt{N}/2$ queries on some input.

Proof. Let $\mathcal{P}$ be the uniform distribution on 1-to-1 functions. Let μ draw f from $\mathcal{P}$ or from $\mathcal{D}$ with probability $1/2$ each. Let A be a deterministic algorithm with $k \leqslant \sqrt{N}/2$ queries, all different, as in the proof of Theorem 31. Its answer is a function of the sequence $T = (f(q_1), \ldots, f(q_k))$ of answers.

Let $t = (a_1, \ldots, a_k)$ have distinct entries. Under $\mathcal{P}$ every answer is uniform among the unused values, so $\mathbb{P}_{\mathcal{P}}(T = t) = \prod_{i=1}^{k} 1/(N - i + 1)$. Under $\mathcal{D}$, let p_i be the probability of a collision at step i, given no collision before and the answers $a_1, \ldots, a_{i-1}$. By the proof of Lemma 33, $p_i \leqslant 2(i-1)/N$. Without a collision the ith answer is uniform among the $N - i + 1$ unused values. By Lemma 36 below,

$$\mathbb{P}_{\mathcal{D}}(T = t) \geqslant \prod_{i=1}^{k} \frac{1 - p_i}{N - i + 1} \geqslant \left(1 - \sum_{i=1}^{k} \frac{2(i-1)}{N}\right) \mathbb{P}_{\mathcal{P}}(T = t)$$
$$= \left(1 - \frac{k(k-1)}{N}\right) \mathbb{P}_{\mathcal{P}}(T = t) \geqslant \frac{3}{4}\, \mathbb{P}_{\mathcal{P}}(T = t),$$

because $k(k-1) \leqslant k^2 \leqslant N/4$.

Let $\mathcal{T}$ be the set of sequences on which A answers "1-to-1", and let $q = \mathbb{P}_{\mathcal{P}}(T \in \mathcal{T})$. Sequences with a repeated entry have probability 0 under $\mathcal{P}$. Summing over $t \in \mathcal{T}$ gives $\mathbb{P}_{\mathcal{D}}(T \in \mathcal{T}) \geqslant \frac{3}{4}q$. So A is correct on an input drawn from μ with probability at most

$$\frac{1}{2}q + \frac{1}{2}\left(1 - \frac{3}{4}q\right) = \frac{1}{2} + \frac{q}{8} \leqslant \frac{5}{8} < \frac{2}{3}.$$

So A errs with probability at least $3/8 > 1/3$ on μ. The claim follows from Lemma 32. □

Lemma 36

For $a_1, \ldots, a_m \in [0, 1]$, $\prod_{i=1}^{m}(1 - a_i) \geqslant 1 - \sum_{i=1}^{m} a_i$.

Proof. We use induction on m, with equality for $m = 1$. Multiplying the hypothesis for $m - 1$ by $1 - a_m \geqslant 0$ gives

$$\prod_{i=1}^{m}(1 - a_i) \geqslant \left(1 - \sum_{i=1}^{m-1} a_i\right)(1 - a_m) = 1 - \sum_{i=1}^{m} a_i + a_m \sum_{i=1}^{m-1} a_i \geqslant 1 - \sum_{i=1}^{m} a_i.$$ □

2 Simon's Algorithm Revisited

In the state Equation (8) just before the measurement, the probability of measuring y in the control register is the sum of the squared amplitudes of all basis states $|y\rangle|z\rangle$ with this y. This is the squared norm of the vector that multiplies $|y\rangle$:

$$\mathbb{P}(y) = \left\| \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{x \cdot y} |f(x)\rangle \right\|_2^2 .$$

2.1 The 1-to-1 Case

Exercise 19. Suppose that f is 1-to-1. What is the probability of measuring a given string y in Simon's circuit, and do these probabilities add up to 1?

Solution. If f is 1-to-1, the 2^n vectors $|f(x)\rangle$ are distinct basis states, so they are orthonormal. The vector multiplying $|y\rangle$ then has 2^n orthogonal components, each of absolute value $1/2^n$. Its squared norm is

$$\mathbb{P}(y) = \sum_{x \in \{0,1\}^n} \left(\frac{1}{2^n}\right)^2 = 2^n \cdot \frac{1}{2^{2n}} = \frac{1}{2^n}.$$

So the measured y is uniformly distributed over $\{0,1\}^n$, and the probabilities add up to $2^n \cdot 1/2^n = 1$. □

2.2 The 2-to-1 Case

If f is 2-to-1 with period $s \neq 0^n$, every y with $s \cdot y = 0$ is measured with probability $1/2^{n-1}$ and every y with $s \cdot y = 1$ with probability 0 (Theorem 28, case 1). There are 2^{n-1} strings with $s \cdot y = 0$, so the probabilities add up to $2^{n-1} \cdot 1/2^{n-1} = 1$.

2.3 Solving for s

Algorithm 1 runs the circuit $n-1$ times. In the 2-to-1 case the measured strings $y_1, \ldots, y_{n-1}$ give the system of $n-1$ linear equations $s \cdot y_1 = 0, \ldots, s \cdot y_{n-1} = 0$ over $\{0,1\}$. If the y_i are linearly independent, the system has rank $n-1$, and Gaussian elimination gives its only nonzero solution s'. The algorithm then tests $f(0^n) = f(s')$ with two classical queries. If f is 2-to-1, then s solves the system, so $s' = s$ and the test is true. If f is 1-to-1, the test is false because $s' \neq 0^n$ (Theorem 29).

Exercise 20. Let V be a subspace of $\{0,1\}^n$ of dimension $d \geqslant 1$, so $|V| = 2^d$, and let $y_1, y_2, \ldots$ be independent uniformly random elements of V. Show that

1. $y_1, \ldots, y_d$ are linearly independent with probability

$$\prod_{i=1}^{d} \left(1 - \frac{1}{2^i}\right) = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{2^3}\right) \cdots \left(1 - \frac{1}{2^d}\right) \geqslant \frac{1}{4},$$

2. $y_1, \ldots, y_{d-1}$ are linearly independent with probability $\prod_{i=2}^{d}(1 - 1/2^i) \geqslant 1/2$.

Deduce that the strings $y_1, \ldots, y_{n-1}$ of Simon's algorithm are linearly independent with probability at least $1/4$ in the 2-to-1 case ($V = \{y \mid s \cdot y = 0\}$, $d = n-1$) and at least $1/2$ in the 1-to-1 case ($V = \{0,1\}^n$, $d = n$).

Proof. Suppose that $y_1, \ldots, y_j$ are linearly independent, $0 \leqslant j < d$. Their span consists of the 2^j vectors $\sum_{i \leqslant j} c_i y_i$ with $c_i \in \{0,1\}$, all different by independence, and it lies inside V. The vector y_{j+1} is uniform over the 2^d elements of V and independent of $y_1, \ldots, y_j$. Then $y_1, \ldots, y_{j+1}$ are linearly independent exactly when y_{j+1} is not in this span, which has probability $1 - 2^j/2^d = 1 - 2^{-(d-j)}$. Multiplying for $j = 0, \ldots, m-1$, the first m vectors are linearly independent with probability

$$\prod_{j=0}^{m-1} \left(1 - \frac{1}{2^{d-j}}\right) = \prod_{i=d-m+1}^{d} \left(1 - \frac{1}{2^i}\right),$$

which gives the two products for $m = d$ and $m = d-1$. By Lemma 36 and the geometric series $\sum_{i \geqslant 2} 2^{-i} = 1/2$,

$$\prod_{i=2}^{d} \left(1 - \frac{1}{2^i}\right) \geqslant 1 - \sum_{i=2}^{d} \frac{1}{2^i} \geqslant \frac{1}{2}, \qquad \prod_{i=1}^{d} \left(1 - \frac{1}{2^i}\right) = \frac{1}{2} \prod_{i=2}^{d} \left(1 - \frac{1}{2^i}\right) \geqslant \frac{1}{4}.$$

In the 2-to-1 case each y_i is uniform over the 2^{n-1} strings with $s \cdot y = 0$ (Theorem 28). They are the solution set of one nontrivial homogeneous linear equation, so they form a subspace of dimension $d = n-1$, and we take $n - 1 = d$ samples. In the 1-to-1 case each y_i is uniform over $\{0,1\}^n$ (Exercise 19), $d = n$, and we take $d - 1$ samples. □

Week: 7 (October 5–9, 2026)

Lecture Date: Tuesday, October 6, 2026

Lecture Topic: Grover's Search Algorithm, Phase Flip, Diffusion Operator, Rotation Analysis

In this lecture, we study *Grover's search algorithm* for unstructured search over $N = 2^n$ inputs. Every Grover iteration is a rotation by the angle 2θ in a two-dimensional plane.

Problem 6 ▶ Unstructured Search

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, find an x with $f(x) = 1$.

Think of f as a Boolean formula and of x as an assignment. In *unique satisfiability* we are promised that there is exactly one x with $f(x) = 1$. Let

$$a = |\{x \mid f(x) = 1\}|, \qquad b = |\{x \mid f(x) \neq 1\}| = N - a$$

be the numbers of solutions and of non-solutions. The search is *unstructured*: we know nothing about f except what the queries tell us. For $a = 1$ every classical algorithm, deterministic or randomized, needs $\Omega(N)$ queries (lectures of September 1 and 3). Grover's algorithm needs $\mathcal{O}\left(\sqrt{N}\right)$ queries, and $\mathcal{O}\left(\sqrt{N/a}\right)$ for a solutions.

1 Grover's Algorithm on Two Qubits

Let $n = 2$, $N = 4$, and suppose that the unique solution is $x^* = 10$. We write a state of the two qubits as its vector of four amplitudes $(a_{00}, a_{01}, a_{10}, a_{11})$. The algorithm has four steps (Figure 15).

1. *Uniform superposition with Hadamard gates:* $H^{\otimes 2}|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$.

2. *Phase flip.* The query $U_f\colon |x\rangle|c\rangle \mapsto |x\rangle|c \oplus f(x)\rangle$ with the target qubit in $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ is the phase flip of the lecture of September 24, $U_f\colon |x\rangle \mapsto (-1)^{f(x)}|x\rangle$. One query gives $\left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right)$.

3. *Grover diffusion operator.* Let $\mu = \frac{1}{N}\sum_x a_x$ be the mean of the amplitudes. The diffusion operator D reflects every amplitude about the mean (*inversion about the mean*):

$$D\colon a_x \longmapsto 2\mu - a_x \qquad \text{for every } x.$$

   Here $\mu = \frac{1}{4}$, so the three non-solutions become $2 \cdot \frac{1}{4} - \frac{1}{2} = 0$ and the solution becomes $2 \cdot \frac{1}{4} + \frac{1}{2} = 1$. The state is $(0, 0, 1, 0) = |10\rangle$, so measuring gives $x^* = 10$ with probability 1 after a single query.

4. In general, *repeat steps 2 and 3* k times, where k is about $\sqrt{N}$ (Section 3).

2 Grover's Algorithm on n Qubits

For n qubits and a unique solution x^* the same steps apply to the N amplitudes. After the Hadamard gates every amplitude is $1/\sqrt{N}$. The phase flip makes the amplitude of x^* equal to $-1/\sqrt{N}$. So the mean is $\mu = (N-2)/(N\sqrt{N}) \approx 1/\sqrt{N}$ and the diffusion operator maps the

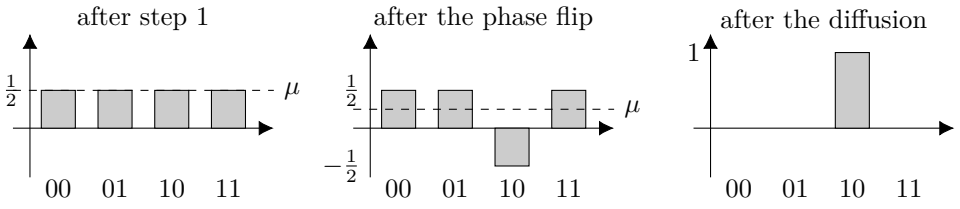


Figure 15: The amplitudes for $N = 4$ and $x^* = 10$ after the Hadamard gates, after the phase flip, and after the inversion about the mean μ.

amplitude of x^* to $2\mu + 1/\sqrt{N} \approx 3/\sqrt{N}$. As long as this amplitude is small, the mean stays close to $1/\sqrt{N}$ and each iteration adds about $2/\sqrt{N}$ to it. So after about $\sqrt{N}$ iterations it is a constant and a measurement finds x^* with constant probability.

2.1 The Operators Z_f, Z_0 and D

The phase flip Z_f. The phase flip

$$Z_f \colon |x\rangle \longmapsto (-1)^{f(x)}|x\rangle$$

is implemented by one query U_f with an extra qubit in $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ (Figure 16). The query leaves the extra qubit unchanged, and a second Hadamard gate returns it to $|1\rangle$.

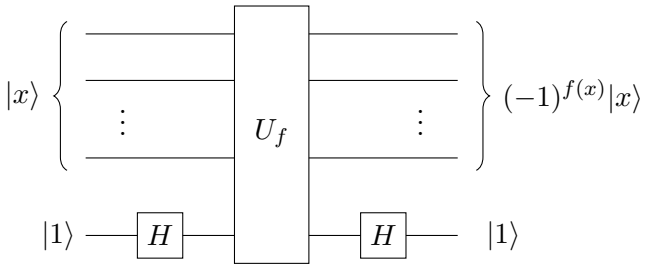


Figure 16: The phase flip Z_f, implemented with one query U_f.

The phase flip Z_0 about 0^n. Let $f_0 \colon \{0,1\}^n \to \{0,1\}$ be the function with $f_0(x) = \bar{x}_1 \wedge \bar{x}_2 \wedge \cdots \wedge \bar{x}_n$, where $\bar{x}_i = \neg x_i$, so $f_0(x) = 1$ only for $x = 0^n$. Its phase flip is

$$Z_0 \colon |x\rangle \longmapsto \begin{cases} -|x\rangle & \text{if } x = 0^n, \\ |x\rangle & \text{otherwise,} \end{cases} \qquad \text{so} \qquad Z_0 = I - 2|0^n\rangle\langle 0^n|.$$

As a matrix, Z_0 is the $N \times N$ diagonal matrix with -1 in the top left corner and 1 elsewhere. It is implemented like Z_f, with an extra qubit in $H|1\rangle$ and the reversible gate $|x\rangle|c\rangle \mapsto |x\rangle|c \oplus (\bar{x}_1 \wedge \cdots \wedge \bar{x}_n)\rangle$ (a Toffoli gate with n negated controls) in place of the black box. It makes no query to f.

The diffusion operator D. Let $|h\rangle = H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$ be the uniform superposition. The diffusion operator is

$$D = -H^{\otimes n} Z_0 H^{\otimes n}.$$

Lemma 37 ▶ Inversion About the Mean

$D = 2|h\rangle\langle h| - I$, and for every state $\sum_x a_x|x\rangle$ with mean amplitude $\mu = \frac{1}{N}\sum_x a_x$,

$$D \sum_{x\in\{0,1\}^n} a_x|x\rangle = \sum_{x\in\{0,1\}^n} (2\mu - a_x)|x\rangle.$$

Proof. Since $H^\dagger = H$ and $H^2 = I$, we have $H^{\otimes n}H^{\otimes n} = I$ and $\langle 0^n|H^{\otimes n} = (H^{\otimes n}|0^n\rangle)^\dagger = \langle h|$, so

$$D = -H^{\otimes n}(I - 2|0^n\rangle\langle 0^n|)H^{\otimes n} = -(I - 2|h\rangle\langle h|) = 2|h\rangle\langle h| - I.$$

For $|\psi\rangle = \sum_x a_x|x\rangle$ we have $\langle h|\psi\rangle = \frac{1}{\sqrt{N}}\sum_x a_x = \sqrt{N}\mu$, so

$$D|\psi\rangle = 2\sqrt{N}\mu|h\rangle - |\psi\rangle = 2\mu\sum_x|x\rangle - \sum_x a_x|x\rangle = \sum_x(2\mu - a_x)|x\rangle. \qquad \square$$

So D is the inversion about the mean of step 3, and it makes no query.

The Grover iteration. One iteration of steps 2 and 3 is the operator

$$G = DZ_f = -H^{\otimes n}Z_0H^{\otimes n}Z_f.$$

Grover's algorithm prepares $|h\rangle = H^{\otimes n}|0^n\rangle$, applies G k times and measures (Figure 17). It makes k queries, one per iteration.

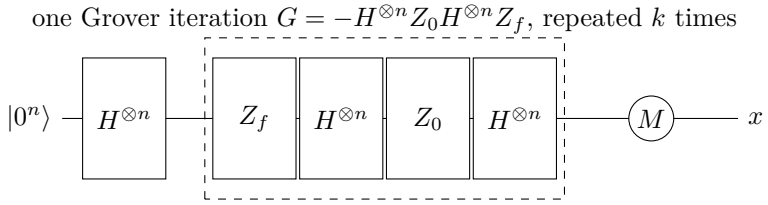


Figure 17: Grover's algorithm on n qubits (the extra qubit used by Z_f and Z_0 is not drawn).

3 Analysis: A Rotation in Two Dimensions

Let f have a solutions, $1 \leqslant a \leqslant N-1$, and let $b = N - a$. Let

$$|A\rangle = \frac{1}{\sqrt{a}}\sum_{\substack{x\in\{0,1\}^n\\ f(x)=1}}|x\rangle, \qquad |B\rangle = \frac{1}{\sqrt{b}}\sum_{\substack{x\in\{0,1\}^n\\ f(x)=0}}|x\rangle$$

be the uniform superpositions of the solutions and of the non-solutions. They are unit vectors, and they are orthogonal since no basis state appears in both. The whole computation stays in the plane they span. The initial state is

$$|h\rangle = \frac{1}{\sqrt{N}}\sum_{x\in\{0,1\}^n}|x\rangle = \frac{1}{\sqrt{N}}\Big(\sqrt{a}|A\rangle + \sqrt{b}|B\rangle\Big) = \sqrt{\frac{a}{N}}|A\rangle + \sqrt{\frac{b}{N}}|B\rangle. \tag{9}$$

Since $\frac{a}{N} + \frac{b}{N} = 1$, there is a unique angle $\theta \in (0, \pi/2)$ with

$$\sin\theta = \sqrt{\frac{a}{N}}, \qquad \cos\theta = \sqrt{\frac{b}{N}}, \qquad \text{so that} \qquad |h\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle.$$

For a unique solution, $\sin\theta = 1/\sqrt{N}$, so $\theta \approx 1/\sqrt{N}$ and $|h\rangle$ is almost $|B\rangle$ (Figure 18).

Lemma 38 ▶ One Grover Iteration

$$G|A\rangle = \Big(1 - \frac{2a}{N}\Big)|A\rangle - \frac{2\sqrt{ab}}{N}|B\rangle = \cos 2\theta|A\rangle - \sin 2\theta|B\rangle,$$
$$G|B\rangle = \frac{2\sqrt{ab}}{N}|A\rangle - \Big(1 - \frac{2b}{N}\Big)|B\rangle = \sin 2\theta|A\rangle + \cos 2\theta|B\rangle.$$

Proof. By the definition of Z_f, $Z_f|A\rangle = -|A\rangle$ and $Z_f|B\rangle = |B\rangle$. By Equation (9) and orthonormality, $\langle h|A\rangle = \sqrt{a/N}$ and $\langle h|B\rangle = \sqrt{b/N}$. With $D = 2|h\rangle\langle h| - I$ (Lemma 37), substituting Equation (9) for $|h\rangle$ in the last step of each line,

$$G|A\rangle = -D|A\rangle = |A\rangle - 2|h\rangle\langle h|A\rangle = |A\rangle - 2\sqrt{\frac{a}{N}}|h\rangle = \Big(1 - \frac{2a}{N}\Big)|A\rangle - \frac{2\sqrt{ab}}{N}|B\rangle,$$
$$G|B\rangle = D|B\rangle = 2|h\rangle\langle h|B\rangle - |B\rangle = 2\sqrt{\frac{b}{N}}|h\rangle - |B\rangle = \frac{2\sqrt{ab}}{N}|A\rangle - \Big(1 - \frac{2b}{N}\Big)|B\rangle.$$

The trigonometric form follows from $\sin^2\theta = a/N$, $\cos^2\theta = b/N$ and the double angle formulas: $\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2a/N$, $\cos 2\theta = 2\cos^2\theta - 1 = -(1 - 2b/N)$ and $\sin 2\theta = 2\sin\theta\cos\theta = 2\sqrt{ab}/N$. □

Since $a + b = N$, both $1 - 2a/N$ and $2b/N - 1$ equal $(b-a)/N$. We write a state $\beta|B\rangle + \alpha|A\rangle$ of the plane as the column vector $(\beta, \alpha)^T$ in the basis $(|B\rangle, |A\rangle)$. Then G acts as the 2×2 matrix

$$G = \begin{pmatrix} \dfrac{b-a}{N} & -\dfrac{2\sqrt{ab}}{N} \\ \dfrac{2\sqrt{ab}}{N} & \dfrac{b-a}{N} \end{pmatrix} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, \tag{10}$$

whose columns are the coordinates of $G|B\rangle$ and $G|A\rangle$. This is the rotation matrix $R_\phi = \left(\begin{smallmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{smallmatrix}\right)$ with $\phi = 2\theta$. So one Grover iteration *rotates* the plane by the angle 2θ, from $|B\rangle$ towards $|A\rangle$.

Exercise 21. Compute $G|h\rangle = \sqrt{a/N}\,G|A\rangle + \sqrt{b/N}\,G|B\rangle$ and $G^k|h\rangle$ for $k \geqslant 0$, and verify that the matrix of G in Equation (10) is the rotation $R_{2\theta}$.

Solution. *The matrix is a rotation.* By Lemma 38 and $(b-a)/N = 1 - 2a/N = 2b/N - 1$, the two columns of the matrix in Equation (10) are the coordinates of $G|B\rangle = \cos 2\theta|B\rangle + \sin 2\theta|A\rangle$ and $G|A\rangle = -\sin 2\theta|B\rangle + \cos 2\theta|A\rangle$, which are the columns of $R_{2\theta}$ (not R_θ).
The state $G|h\rangle$. By Equation (9), $|h\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle$. By Lemma 38 and the addition formulas for $\sin(\theta + 2\theta)$ and $\cos(\theta + 2\theta)$,

$$\begin{aligned} G|h\rangle &= \sin\theta(\cos 2\theta|A\rangle - \sin 2\theta|B\rangle) + \cos\theta(\sin 2\theta|A\rangle + \cos 2\theta|B\rangle) \\ &= \sin 3\theta|A\rangle + \cos 3\theta|B\rangle. \end{aligned}$$

With $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and $4b/N - 3 = 1 - 4a/N$, this is

$$G|h\rangle = \sqrt{\frac{a}{N}}\Big(3 - \frac{4a}{N}\Big)|A\rangle + \sqrt{\frac{b}{N}}\Big(1 - \frac{4a}{N}\Big)|B\rangle.$$

The state starts at the angle θ from $|B\rangle$ and the iteration adds 2θ (Figure 18).

The state $G^k|h\rangle$. The same computation with any angle β in place of θ gives $G(\sin\beta|A\rangle + \cos\beta|B\rangle) = \sin(\beta+2\theta)|A\rangle + \cos(\beta+2\theta)|B\rangle$. So by induction on k, starting from the angle θ of $|h\rangle$,

$$G^k|h\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle.$$

□

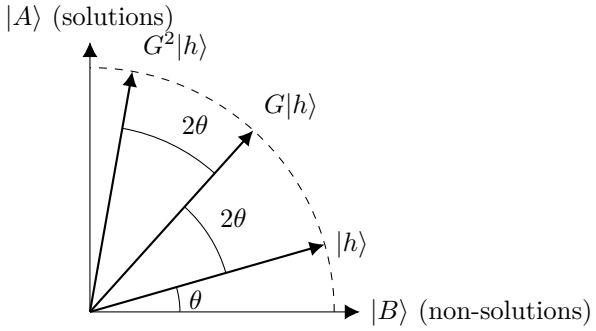


Figure 18: Grover's algorithm in the plane spanned by $|A\rangle$ and $|B\rangle$ for $\theta = 16°$.

The number of iterations is computed in the next lecture (Section 2).

Week: 7 (October 5–9, 2026)

Lecture Date: Thursday, October 8, 2026

Lecture Topic: Grover's Algorithm as Two Reflections, Number of Iterations, Random Number of Iterations, Quantum Amplitude Amplification

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In this lecture, we finish the analysis of *Grover's search algorithm*: one iteration is two *reflections*, which make a rotation. We then choose the number of iterations at *random* and generalize the algorithm to *quantum amplitude amplification*.

Recap. We keep the notation of the last lecture (a, $b = N - a$ with $1 \leqslant a \leqslant N-1$, $|A\rangle$, $|B\rangle$, Z_f, Z_0, D, $|h\rangle$ and the Grover iteration $G = DZ_f = -H^{\otimes n}Z_0H^{\otimes n}Z_f$; Figure 17). By Equation (9), $|h\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle$ with $\theta = \sin^{-1}\left(\sqrt{a/N}\right) \in (0, \pi/2)$. We write a unit vector of the plane spanned by $|A\rangle$ and $|B\rangle$ with real coefficients as $\sin\alpha|A\rangle + \cos\alpha|B\rangle$ and call α its *angle* (measured from $|B\rangle$ towards $|A\rangle$). So $|h\rangle$ has angle θ, and a measurement of the state with angle α gives a solution with probability $\sin^2\alpha$.

1 A Grover Iteration Is Two Reflections

Since $Z_f|A\rangle = -|A\rangle$ and $Z_f|B\rangle = |B\rangle$ (proof of Lemma 38), $Z_f(\sin\alpha|A\rangle + \cos\alpha|B\rangle) = \sin(-\alpha)|A\rangle + \cos(-\alpha)|B\rangle$. So Z_f is the reflection of the plane about the line of $|B\rangle$.

Let $|h^\perp\rangle = \cos\theta|A\rangle - \sin\theta|B\rangle$. It is a unit vector of the plane with $\langle h|h^\perp\rangle = \sin\theta\cos\theta - \cos\theta\sin\theta = 0$, so $|h\rangle$ and $|h^\perp\rangle$ are a second orthonormal basis of the plane (Figure 19).

Exercise 22. Show that $D = 2|h\rangle\langle h| - I$, and that for every state $|\phi\rangle = c|h\rangle + d|h^\perp\rangle$ of

the plane,

$$D|\phi\rangle = c|h\rangle - d|h^\perp\rangle.$$

Solution. The first claim is Lemma 37: $H^{\otimes n}H^{\otimes n} = I$ and $\langle 0^n|H^{\otimes n} = \langle h|$ give

$$D = -H^{\otimes n}(I - 2|0^n\rangle\langle 0^n|)H^{\otimes n} = 2|h\rangle\langle h| - I.$$

Since $\langle h|h\rangle = 1$ and $\langle h|h^\perp\rangle = 0$, we have $\langle h|\phi\rangle = c$, so

$$D|\phi\rangle = 2|h\rangle\langle h|\phi\rangle - |\phi\rangle = 2c|h\rangle - c|h\rangle - d|h^\perp\rangle = c|h\rangle - d|h^\perp\rangle.$$

So D is the reflection of the plane about the line of $|h\rangle$. □

The sign of G. In class the Grover iteration was also written as $G = (H^{\otimes n}Z_0H^{\otimes n})(-Z_f)$, which is the same operator as $DZ_f = (-H^{\otimes n}Z_0H^{\otimes n})Z_f$ because the minus sign can be moved from one factor to the other. The form DZ_f is the reflection about $|B\rangle$ followed by the reflection about $|h\rangle$. In the other form, $-Z_f$ negates the $|B\rangle$ component and $H^{\otimes n}Z_0H^{\otimes n} = I - 2|h\rangle\langle h|$ negates the $|h\rangle$ component, so it is the reflection about $|A\rangle$ followed by the reflection about $|h^\perp\rangle$. Both pairs of lines make the angle θ.

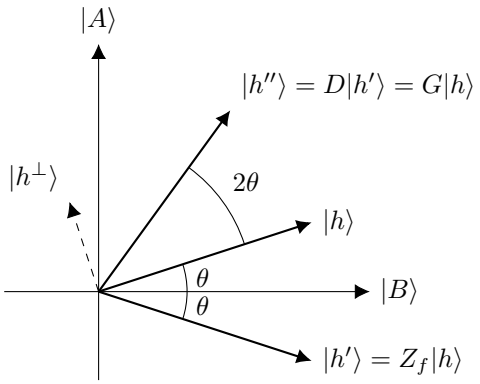


Figure 19: One Grover iteration applied to $|h\rangle$: Z_f reflects about the line of $|B\rangle$, then D reflects about the line of $|h\rangle$.

1.1 Two Reflections Make a Rotation

In Figure 19 the first reflection takes the angle θ to $-\theta$ and the second, about the line at angle θ, takes $-\theta$ to 3θ.

Lemma 39 ► Two Reflections Make a Rotation

Let $|A\rangle$ and $|B\rangle$ be orthonormal vectors, let θ be a real number and $|\psi\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle$. Let S be a linear operator with $S|A\rangle = -|A\rangle$ and $S|B\rangle = |B\rangle$, and let $T = 2|\psi\rangle\langle\psi| - I$. Then for every real number β,

$$TS(\sin\beta|A\rangle + \cos\beta|B\rangle) = \sin(\beta + 2\theta)|A\rangle + \cos(\beta + 2\theta)|B\rangle,$$

and so, for every integer $k \geqslant 0$,

$$(TS)^k|\psi\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle.$$

Proof. By linearity, $S(\sin\beta|A\rangle + \cos\beta|B\rangle) = -\sin\beta|A\rangle + \cos\beta|B\rangle = \sin\gamma|A\rangle + \cos\gamma|B\rangle =: |\chi\rangle$ with $\gamma = -\beta$. By orthonormality, $\langle\psi|\chi\rangle = \sin\theta\sin\gamma + \cos\theta\cos\gamma = \cos(\gamma-\theta)$, so

$$\begin{aligned} T|\chi\rangle &= 2\cos(\gamma-\theta)|\psi\rangle - |\chi\rangle \\ &= (2\cos(\gamma-\theta)\sin\theta - \sin\gamma)|A\rangle + (2\cos(\gamma-\theta)\cos\theta - \cos\gamma)|B\rangle. \end{aligned}$$

By the product formulas $2\cos u\sin v = \sin(u+v) - \sin(u-v)$ and $2\cos u\cos v = \cos(u+v) + \cos(u-v)$ with $u = \gamma-\theta$ and $v = \theta$, the coefficients are $-\sin(\gamma-2\theta) = \sin(2\theta-\gamma)$ and $\cos(\gamma-2\theta) = \cos(2\theta-\gamma)$. So T sends the angle γ to $2\theta-\gamma$, and TS sends β to $2\theta-(-\beta) = \beta+2\theta$. This is the first claim. The state $|\psi\rangle$ has the angle θ and each application of TS adds 2θ, so induction on k gives the angle $(2k+1)\theta$ for $(TS)^k|\psi\rangle$. □

For Grover's algorithm we take $S = Z_f$, $|\psi\rangle = |h\rangle$ and $T = D = 2|h\rangle\langle h| - I$ (Exercise 22), so that $TS = DZ_f = G$:

Corollary 40 ▶ The State After k Iterations

Every Grover iteration rotates the plane by 2θ from $|B\rangle$ towards $|A\rangle$, and the state after k iterations is

$$|h^k\rangle = G^k|h\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle.$$

A measurement of $|h^k\rangle$ gives a solution with probability $\sin^2((2k+1)\theta)$.

1.2 The Matrix of a Grover Iteration

Let M be the matrix of the Grover iteration in the basis $(|B\rangle, |A\rangle)$ (Equation (10)).

Exercise 23. Show that

$$M = \begin{pmatrix} \sqrt{b/N} & -\sqrt{a/N} \\ \sqrt{a/N} & \sqrt{b/N} \end{pmatrix}^2 = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^2,$$

that is, the Grover iteration is the rotation by θ applied twice.

Solution. Let $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ with $\cos\theta = \sqrt{b/N}$ and $\sin\theta = \sqrt{a/N}$. Multiplying,

$$R_\theta^2 = \begin{pmatrix} \cos^2\theta - \sin^2\theta & -2\sin\theta\cos\theta \\ 2\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{pmatrix} = \begin{pmatrix} \dfrac{b-a}{N} & -\dfrac{2\sqrt{ab}}{N} \\ \dfrac{2\sqrt{ab}}{N} & \dfrac{b-a}{N} \end{pmatrix} = M,$$

and by the double angle formulas the middle matrix is $R_{2\theta}$. So $M^k = R_{2k\theta}$. This is Corollary 40 in matrix form: $|h\rangle$ is the first column $(\cos\theta, \sin\theta)^T$ of R_θ, so $M^k|h\rangle$ is the first column of $R_{(2k+1)\theta}$. □

2 The Number of Iterations

By Corollary 40 we want $(2k+1)\theta \approx \pi/2$, so $k \approx \frac{\pi}{4\theta} - \frac{1}{2}$. With $\theta \approx \sin\theta = \sqrt{a/N}$ we take

$$k = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{a}} \right\rfloor, \qquad \text{for a unique solution} \quad k = \left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor.$$

Exercise 24. Let f have a solutions, $1 \leqslant a \leqslant N/2$, and let $\kappa = \frac{\pi}{4}\sqrt{N/a}$. Show that Grover's algorithm with $k = \lfloor \kappa \rfloor$ iterations makes $k \geqslant 1$ queries and outputs a solution with probability at least $1 - 2a/N$ (at least $1 - 2/N$ for a unique solution).

Proof. Let $t = \frac{\pi}{4\theta} - \frac{1}{2}$, so that $(2t+1)\theta = \pi/2$ and $(2k+1)\theta = \frac{\pi}{2} + 2(k-t)\theta$ for every integer k. By Corollary 40, $\sin(\pi/2 + x) = \cos x$ and $|\sin x| \leqslant |x|$, the success probability is

$$\sin^2((2k+1)\theta) = \cos^2(2(k-t)\theta) = 1 - \sin^2(2(k-t)\theta) \geqslant 1 - 4(k-t)^2\theta^2. \tag{11}$$

Two facts about θ. Since $\sin\theta = \sqrt{a/N} \leqslant 1/\sqrt{2}$, we have $0 < \theta \leqslant \pi/4$, so $\theta < 0.79$ and $\theta^2 < 0.62$. We use $\theta - \theta^3/6 \leqslant \sin\theta \leqslant \theta$ for $\theta \geqslant 0$. The lower bound holds because $g(x) = \sin x - x + x^3/6$ has $g(0) = 0$ and $g'(x) = \cos x - 1 + x^2/2 \geqslant 0$, as $1 - \cos x = 2\sin^2(x/2) \leqslant x^2/2$. So $\sin\theta \geqslant \theta(1 - \theta^2/6)$ with $1 - \theta^2/6 > 1 - 0.62/6 > 0.896$. First, as $0.896^2 > 0.8$,

$$\theta^2 \leqslant \frac{\sin^2\theta}{(1-\theta^2/6)^2} < \frac{\sin^2\theta}{0.896^2} < \frac{5}{4}\sin^2\theta = \frac{5a}{4N}. \tag{12}$$

Second, $\kappa = \frac{\pi}{4\sin\theta}$, and

$$0 \leqslant \frac{1}{\sin\theta} - \frac{1}{\theta} \leqslant \frac{1}{\theta(1-\theta^2/6)} - \frac{1}{\theta} = \frac{\theta}{6-\theta^2} < \frac{0.79}{5.38} < 0.15,$$

so, multiplying by $\pi/4 < 0.79$, we get $0 \leqslant \kappa - \pi/4\theta < 0.79 \cdot 0.15 < 1/8$. This gives $\frac{1}{2} \leqslant \kappa - t < \frac{5}{8}$.
The bound. For $k = \lfloor \kappa \rfloor$ we have $-1 < k - \kappa \leqslant 0$, so $-\frac{1}{2} < k - t = (k - \kappa) + (\kappa - t) < \frac{5}{8}$ and $(k-t)^2 < 25/64$. By Equations (11) and (12) the success probability is at least

$$1 - 4 \cdot \frac{25}{64} \cdot \frac{5a}{4N} = 1 - \frac{125}{64} \cdot \frac{a}{N} \geqslant 1 - \frac{2a}{N}.$$

Also $\kappa \geqslant \frac{\pi}{4}\sqrt{2} > 1$ because $N/a \geqslant 2$, so $k \geqslant 1$. □

With $\lceil \kappa \rceil$ instead of $\lfloor \kappa \rfloor$, the bound fails in general. For $a = N/4$ we have $\theta = \pi/6$ and $\kappa = \pi/2 \approx 1.57$. Then $\lfloor \kappa \rfloor = 1$ iteration gives the angle $\pi/2$ and success probability 1, but $\lceil \kappa \rceil = 2$ iterations overshoot to $5\pi/6$ and success probability $1/4$.

3 A Random Number of Iterations

The number of iterations $\lfloor \frac{\pi}{4}\sqrt{N/a} \rfloor$ depends on the number a of solutions, which we may not know. So we choose k *at random*. Since $\sqrt{N}$ need not be an integer, the largest value is $\lfloor 2\sqrt{N} \rfloor$.

Exercise 25. Let $N = 2^n$, let f have a solutions, $1 \leqslant a \leqslant N/2$, and let $K = \lfloor 2\sqrt{N} \rfloor$. Choose k uniformly at random from $\{1, \ldots, K\}$, make k Grover iterations on $|h\rangle$ and measure. Prove that the probability of measuring a solution is

$$\mathbb{P}(\text{success}) = \frac{1}{2} - \frac{\sin(4(K+1)\theta) - \sin 4\theta}{4K\sin 2\theta} \geqslant \frac{1}{2} - \frac{1}{4K\sin 2\theta} \geqslant \frac{3}{8}, \tag{13}$$

so that $\mathbb{P}(\text{success}) = \Theta(1)$ with at most $2\sqrt{N} = \mathcal{O}\left(\sqrt{N}\right)$ queries.

Proof. The average. Given k, the success probability is $\sin^2((2k+1)\theta)$ by Corollary 40, and each value has probability $1/K$. By the law of total probability and $\sin^2 x = \frac{1}{2}(1-\cos 2x)$,

$$\mathbb{P}(\text{success}) = \frac{1}{K}\sum_{k=1}^{K}\sin^2((2k+1)\theta) = \frac{1}{2} - \frac{1}{2K}\sum_{k=1}^{K}\cos((4k+2)\theta).$$

By the product formula $2\cos u \sin v = \sin(u+v) - \sin(u-v)$ with $u = (4k+2)\theta$ and $v = 2\theta$, the sum telescopes:

$$2\sin 2\theta \sum_{k=1}^{K}\cos((4k+2)\theta) = \sum_{k=1}^{K}(\sin(4(k+1)\theta) - \sin(4k\theta)) = \sin(4(K+1)\theta) - \sin 4\theta.$$

As $\sin\theta = \sqrt{a/N} \leqslant 1/\sqrt{2}$, we have $0 < \theta \leqslant \pi/4$ and $\sin 2\theta > 0$, so dividing by $2\sin 2\theta$ gives the equality in Equation (13).
The lower bound. $\sin(4(K+1)\theta) \leqslant 1$, and $\sin 4\theta \geqslant 0$ because $0 < 4\theta \leqslant \pi$. So the numerator is at most 1, which gives the first inequality in Equation (13). Since $b = N - a \geqslant N/2$,

$$\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\sqrt{ab}}{N} \geqslant \frac{2\sqrt{a\cdot N/2}}{N} = \sqrt{\frac{2a}{N}}.$$

Next, $\lfloor 2\sqrt{N}\rfloor \geqslant \sqrt{2N}$ for $N = 2^n$. If n is even, $2\sqrt{N}$ is an integer. If n is odd, $\sqrt{2N} = 2^{(n+1)/2}$ is an integer with $\sqrt{2N} \leqslant 2\sqrt{N}$, so it is at most the floor. So $K\sin 2\theta \geqslant \sqrt{2N}\cdot\sqrt{2a/N} = 2\sqrt{a} \geqslant 2$ and $\mathbb{P}(\text{success}) \geqslant \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$. A probability is at most 1, so $\mathbb{P}(\text{success}) = \Theta(1)$. The number of queries is $k \leqslant K \leqslant 2\sqrt{N}$. □

For the set $\left\{1,\dots,\lfloor\sqrt{N}\rfloor\right\}$ the same proof with $K = \lfloor\sqrt{N}\rfloor \geqslant \sqrt{N/2}$ gives $\mathbb{P}(\text{success}) \geqslant 1/4$.

4 Quantum Amplitude Amplification

Let $\mathcal{A}$ be any quantum algorithm without measurements (a unitary operator on n qubits), and let $f\colon \{0,1\}^n \to \{0,1\}$ tell us which outcomes x are *good* ($f(x) = 1$) and which are *bad* ($f(x) = 0$). Write $\mathcal{A}|0^n\rangle = \sum_x c_x|x\rangle$, let $p = \sum_{x|f(x)=1}|c_x|^2$ be the probability that a measurement of $\mathcal{A}|0^n\rangle$ gives a good outcome, and assume $0 < p < 1$. With the unit vectors

$$|G\rangle = \frac{1}{\sqrt{p}}\sum_{x|f(x)=1} c_x|x\rangle, \qquad |B\rangle = \frac{1}{\sqrt{1-p}}\sum_{x|f(x)=0} c_x|x\rangle,$$

which are orthogonal because no basis state appears in both, the state that $\mathcal{A}$ produces is $|\phi\rangle = \mathcal{A}|0^n\rangle = \sqrt{p}\,|G\rangle + \sqrt{1-p}\,|B\rangle$. Here $|G\rangle$ is the good state, unrelated to the Grover iteration G. *Classically* we would run $\mathcal{A}$, measure, check the outcome with f, and repeat until the outcome is good. The number of runs is geometric with mean $1/p$.

Exercise 26. Let $\theta \in (0, \pi/2)$ with $\sin\theta = \sqrt{p}$, and let

$$Q = -\mathcal{A}Z_0\mathcal{A}^{-1}Z_f.$$

Show that $Q^k|\phi\rangle = \sin((2k+1)\theta)|G\rangle + \cos((2k+1)\theta)|B\rangle$ for every $k \geqslant 0$. Show that for $k = \lfloor\pi/4\theta\rfloor$, a measurement of $Q^k|\phi\rangle$ gives a good outcome with probability at least $\max(p, 1-p) \geqslant 1/2$, and that computing $Q^k\mathcal{A}|0^n\rangle$ takes $2k+1 \leqslant \frac{\pi}{2\sqrt{p}} + 1 = \mathcal{O}(1/\sqrt{p})$ applications of $\mathcal{A}$ or $\mathcal{A}^{-1}$ and k queries to f.

Proof. The two reflections. Z_f negates every good basis state and fixes every bad one, so $Z_f|G\rangle = -|G\rangle$ and $Z_f|B\rangle = |B\rangle$. Since $\mathcal{A}$ is unitary, $\mathcal{A}^{-1} = \mathcal{A}^\dagger$, so $\mathcal{A}I\mathcal{A}^{-1} = I$ and $\langle 0^n|\mathcal{A}^{-1} = (\mathcal{A}|0^n\rangle)^\dagger = \langle\phi|$. This gives

$$-\mathcal{A}Z_0\mathcal{A}^{-1} = -\mathcal{A}(I - 2|0^n\rangle\langle 0^n|)\mathcal{A}^{-1} = -(I - 2|\phi\rangle\langle\phi|) = 2|\phi\rangle\langle\phi| - I.$$

So $Q = TS$ with $S = Z_f$ and $T = 2|\phi\rangle\langle\phi| - I$. Also $|\phi\rangle = \sin\theta|G\rangle + \cos\theta|B\rangle$ because $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1-p}$. Lemma 39, applied to the orthonormal vectors $|G\rangle$ and $|B\rangle$ and to $|\psi\rangle = |\phi\rangle$, gives the formula for $Q^k|\phi\rangle$.

The success probability. $|G\rangle$ contains only good basis states and $|B\rangle$ only bad ones, so a measurement of $Q^k|\phi\rangle$ gives a good outcome with probability $\sin^2((2k+1)\theta)$. For $k = \lfloor \pi/4\theta \rfloor$ we have $\pi/4\theta - 1 < k \leqslant \pi/4\theta$, so $\frac{\pi}{2} - \theta < (2k+1)\theta \leqslant \frac{\pi}{2} + \theta$. On the interval $[\pi/2 - \theta, \pi/2 + \theta] \subseteq [0, \pi]$ the sine is at least its value at the endpoints, $\sin(\pi/2 \pm \theta) = \cos\theta$, so the probability is at least $\cos^2\theta = 1 - p$. It is also at least p: if $p \leqslant 1/2$ then $1 - p \geqslant p$, and if $p > 1/2$ then $\theta > \pi/4$, so $k = 0$ and the probability is $\sin^2\theta = p$.

The cost. One application of Q uses $\mathcal{A}$ once, $\mathcal{A}^{-1}$ once and Z_f once (one query to f), and Z_0 uses neither. Preparing $|\phi\rangle$ uses $\mathcal{A}$ once more, so the total is $2k + 1$ applications. Also $k \leqslant \frac{\pi}{4\theta} \leqslant \frac{\pi}{4\sin\theta} = \frac{\pi}{4\sqrt{p}}$ because $\sin\theta \leqslant \theta$. □